

Selling Supplemental Information ^{*}

Arlindo Skënderaj [†]

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Abstract

I consider an environment in which a decision maker (e.g., a monopolist) faces uncertainty about a payoff-relevant state (e.g., consumer valuations). The decision maker privately owns a partially informative signal about the state and can choose to purchase supplemental information from a data broker. I study the data broker’s optimal selling mechanism, which involves screening over the space of all signals. The data broker can offer signals that are arbitrarily correlated with the decision maker’s private signal, a feature that plays a central role in the design of the optimal menu. When the decision maker chooses between two actions, the data broker extracts the full surplus for any type space. With more actions and states, full surplus extraction is possible whenever all suboptimal actions yield the same payoff in any given state. In both cases, a menu that assigns each type a *confirm-or-reveal* signal is optimal, and it is independent of the data broker’s prior. I also identify environments in which full surplus extraction fails and characterize the optimal menu when there are two decision maker types.

Keywords: Data, Information Design, Mechanism Design, Multidimensional Screening, Selling Information

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[†]Brown University, Department of Economics, arlindo_skenderaj@brown.edu.

1 Introduction

Firms are increasingly relying on data to make better decisions. One particularly prominent channel for this is the use of consumer data for price discrimination and product targeting. Another is the use of credit data to evaluate loan applications. Although firms collect first-party data through direct interactions with their consumers, this information is often incomplete. To fill these gaps, they turn to data brokers (e.g., Experian, LiveRamp, Nielsen), who can provide them with any supplemental information but do not observe the information firms have. The value of new data depends on the data that a decision maker already has. Hence, data brokers must account for what the decision maker already owns to tailor the supplemental data to her existing dataset.

I develop a framework to analyze the sale of supplemental information to a privately informed decision maker. The decision maker faces uncertainty about the true state of the world and, consequently, about which action to take. She privately holds partial information in the form of a signal à la [Gentzkow and Kamenica \(2017\)](#), which defines her type. To improve her decision, she can purchase supplemental information from a data broker. The data broker does not observe the decision maker's type, and can choose to sell any signal. In particular, he can sell signals that are arbitrarily correlated with the decision maker's private signal.

To illustrate the framework, suppose a decision maker is a monopolist who sells concert tickets to students and professionals. She faces a market of unit mass consisting of 40% students and 60% professionals, with a willingness to pay \$50 and \$100, respectively. If she had information that fully distinguished students from professionals, she would engage in first-degree price discrimination and earn an expected profit of \$80 by charging \$50 to students and \$100 to professionals.

Assume now that the monopolist does not own a fully revealing dataset, but is instead one of two types. The Type I decision maker is fully uninformed, charges the monopoly price of \$100, and earns an expected profit of \$60. The Type II decision maker owns a residential dataset (see [Figure 1](#)), which indicates whether a consumer lives in a house or an apartment. Among consumers living in a house, the proportion of students is

$$P(\text{Student} \mid \text{House}) = \frac{P(\text{House} \mid \text{Student}) \cdot P(\text{Student})}{P(\text{House})} = \frac{0.4 \cdot 0.3}{0.4 \cdot 0.3 + 0.6 \cdot 0.8} = 20\%,$$

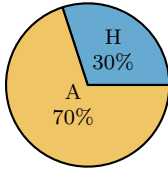
and the remaining 80% are professionals; among those living in an apartment, the proportion

of students is

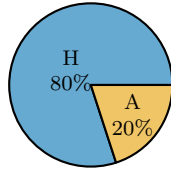
$$P(\text{Student} \mid \text{Apt}) = \frac{P(\text{Apt} \mid \text{Student}) \cdot P(\text{Student})}{P(\text{Apt})} = \frac{0.4 \cdot 0.7}{0.4 \cdot 0.7 + 0.6 \cdot 0.2} = 70\%,$$

and the remaining 30% are professionals. It is therefore optimal for Type II to charge \$100 to consumers in houses and \$50 to those in apartments, earning a profit of

$$0.6 \cdot (0.8 \cdot \$100) + 0.4 \cdot \$50 = \$68.$$

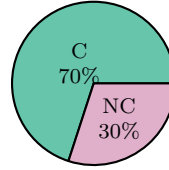


Student
40%

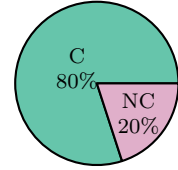


Professional
60%

Figure 1: Residential data
(H = house, A = apartment).



Student
40%



Professional
60%

Figure 2: Vehicle registration data
(C = car owner, NC = non-owner).

Consider now the data broker's problem. The data broker can sell any data to the monopolist but is uncertain about her type, i.e., whether or not the monopolist owns the residential data. One option is to sell the fully revealing data, which would yield the monopolist a profit of \$80. If the monopolist does not own the residential dataset, she is willing to pay up to \$20 for the fully revealing data; if she already owns it, she is willing to pay up to \$12. However, this is not the most profitable way for the data broker to sell data, because that will require either excluding Type II, or selling to both types at a price of \$12.

The data broker can increase profits by offering a partially informative signal, here depicted in the form of vehicle registration data (see Figure 2). Consider how each type of the monopolist would value this dataset. Type I, who originally owns nothing, would have access only to this dataset, under which it is optimal to charge \$100 to both car owners and non-owners; hence, owning vehicle registration data alone is of no value to this type. Type II would now have access to two datasets, the residential and the vehicle registration data. In particular, combining these two datasets fully reveals the consumer's valuation: each combination of residential status and vehicle ownership determines whether an individual is a student or a professional. With both datasets together, the monopolist can engage in perfect price discrimination and earn a profit of \$80. The data broker can therefore extract all rents from the monopolist by selling the fully revealing data for \$20 and the vehicle registration data at \$12. Moreover, this construction is independent of the data broker's prior. Section 4

generalizes this result to any type space.

The general problem faced by the data broker consists of screening over all signals, which can be challenging for two main reasons. First, this is a multidimensional screening problem. Second, the type space is not totally ordered due to both horizontal and vertical differentiation across signals. The value of any supplemental signal depends on the signal the decision maker already has; a particular signal might be highly valuable to one type but entirely useless to another. Much of the existing mechanism design literature relies on ranked type spaces where the single-crossing or increasing differences property holds. In this setting, the single-crossing property does not hold.

The main result shows that the data broker extracts the full surplus in a large class of cases, attained by assigning to each type a *confirm-or-reveal* signal: one message confirms when the interim action is optimal, and the remaining messages reveal the correct action otherwise. Such a menu is optimal and it is independent of data broker's prior over the decision maker's type space.

I first analyze a setting in which the decision maker chooses between two actions, and I provide an explicit construction of the optimal menu. With binary actions, a *confirm-or-reveal* signal is interpreted as a switch/don't-switch recommendation relative to the interim action induced by the type's own signal. A key property of this construction is that the incremental value of such a signal is maximized when paired with the base signal for which it was designed. This ensures incentive compatibility and allows the data broker to extract the full surplus. The intuition is as follows. A deviating type who purchases another type's recommendation faces a recommendation defined relative to that other type's interim action, and can choose either to obey or disobey it. If she obeys, she switches actions whenever the other type switches. The other type always benefits from switching, whereas the deviating type sometimes benefits and sometimes overturns an optimal action. Since the benefit from being corrected in any given state is the same for both types, her gain from obeying is no greater. If she disobeys, the outcome is equivalent to first switching the default action and then obeying. Since switching the default imposes a loss, and the benefit from obeying cannot exceed that of the other type, disobeying is never better.

Then I consider the general problem allowing for many actions and many states. Under this level of generality, I first find that full surplus extraction is not as easily obtained. Indeed, outside of the binary action case, the marginal benefit of being corrected depends on both the action and the state, and therefore, differs across types. As a result, the data broker may not always be able to design a menu that gives each type an efficient allocation without giving away some rents (see Section 4.2 for an example).

Nevertheless, I find conditions under which the data broker extracts the full surplus:

whenever the marginal gain from being corrected depends only on the state. Many decision problems have this form, in which the decision maker must identify the right action to take and every other action is equally costly: routing a package to the correct address, or serving the ad for the product a user actually wants. Proposition 4 shows that the data broker extracts the full surplus whenever all suboptimal actions yield the same payoff in any given state, and provides an explicit characterization of the optimal menu. The menu assigns each decision maker a *confirm-or-reveal* signal that can be interpreted as an action recommendation that either confirms an optimal action or, whenever the interim action is suboptimal, identifies the optimal one by *ruling out* actions the decision maker should not otherwise take. The switch/don't-switch recommendation is a special case in the binary actions setting.

When full surplus extraction fails, I characterize the optimal menu with two types. As in Haghpanah and Siegel (2025), the high type is assigned a signal that induces her to always take the optimal action, and the low type does not get any rent. The data broker then faces a quality-degradation tradeoff: better information for the low type raises revenue but also raises the rent needed to keep the high type from mimicking her. I provide conditions under which the low type is never excluded, even when her frequency is close to zero. It holds whenever there is an event where the low type takes a suboptimal action while the high type's action is uniquely optimal, in which case the data broker can design a signal that benefits the low type while leaving the high type's action unchanged.

The structure of the paper is as follows: In Section 2, I provide related literature. In Section 3, I introduce the model, and in Section 4, I conduct the main analysis. The conclusion is presented in Section 5. Proofs are provided in the Appendix.

2 Related Literature

This paper relates to the literature on selling information to a privately informed decision maker. It departs from existing work in that the decision maker's private information is itself a signal, and the data broker can sell signals that are arbitrarily correlated with it. The closest paper is Bergemann et al. (2018). They study the optimal sale of Blackwell experiments to a decision maker whose private information is an interim belief about the state. In contrast, in this paper, the decision maker's type is a signal rather than an interim belief, and the data broker sells supplemental information that complements this signal. This captures situations where the decision maker makes frequent decisions and purchases information before her private message is realized.

A broader literature studies the optimal sale of information¹. Seminal work includes [Admati and Pfleiderer \(1986\)](#), [Admati and Pfleiderer \(1990\)](#), and [Allen \(1990\)](#), which study the sale of information to traders in financial markets. [Eső and Szentes \(2007b\)](#) study a consultant who sells advice to a client. There the client’s type is a scalar belief, and the payment depend on realized signal and the client’s project’s outcome, whereas in my paper each type pays a fixed price before the state is realized. More recent work studies environments in which a data broker sells information to an agent facing a decision problem under uncertainty. In [Babaioff et al. \(2012\)](#) data broker sells information after the state is realized and to a decision maker with private payoff function. In [Liu et al. \(2021\)](#), the decision maker chooses between a passive and an active action, where the payoff depends on the state only under the active action, reducing the problem to one dimension. [Yang \(2022\)](#) studies the optimal selling mechanism of a data broker who sells a signal to a monopolist with a privately known constant marginal cost. [Segura-Rodriguez \(2022\)](#) studies a data broker who sells information about consumers to a monopolist who wants to predict one characteristic; which characteristic she wants to predict and how much she values the information are both private. In these papers, the decision maker’s private type is a payoff parameter rather than an information structure. Here, the data broker can tailor to each type a signal that is most valuable precisely when combined with the signal she already owns, and extracts the full surplus in a large class of environments, for any type space and independently of his prior.

Full surplus extraction has been studied extensively in mechanism design, beginning with [Crémer and McLean \(1985, 1988\)](#) and extended to continuous type spaces (for arbitrary small rents) by [McAfee and Reny \(1992\)](#). These papers exploit correlation across multiple agents’ private types: each agent’s payment is indexed by the others’ reports, and under a full-rank condition on conditional beliefs the resulting lotteries leave every type zero rent. [Riordan and Sappington \(1988\)](#) attains the same outcome in a single-agent setting by conditioning transfers on a public ex post signal correlated with the agent’s type. More recently, in [Krähmer \(2020\)](#) the full surplus is extracted by privately randomizing over information structures and penalizing reports inconsistent with the realized structure. [Hoxha \(2024\)](#) studies a data broker who sells Blackwell experiments to a decision maker who is privately informed about the state that she cares about and the value of resolving it. There, data broker extracts the full surplus when the correlation across states is small. In this paper, the data broker offers a deterministic menu of signals at fixed prices, each type’s transfer depends only on its own report, and incentive compatibility arises from how the data broker correlates the supplemental signal with the one the decision maker already owns.

Representing information as a signal à la [Gentzkow and Kamenica \(2017\)](#) captures how

¹For an overview of markets for information, see [Bergemann and Bonatti \(2019\)](#).

observations from different sources of information are correlated. This allows for a well-defined operation of combining signals and analyzing their joint informational content, capturing how supplemental data can be appended to existing data. [Börger et al. \(2013\)](#) study when two signals are complements or substitutes in every decision problem. The optimal menu here exploits complementarity in a fixed decision problem, offering each type a signal that is most valuable precisely when paired with her own.

On the literature of persuasion without transfers, [Kolotilin et al. \(2017\)](#), [Kolotilin \(2018\)](#), [Candogan and Strack \(2023\)](#), [Ichihashi and Smolin \(2025\)](#), and [Zhu \(2023\)](#) analyze optimal mechanisms without transfers for providing information to privately informed agents. Other studies have analyzed information transmission from experts to customers (see [Milgrom and Roberts \(1986\)](#), [Pitchik and Schotter \(1987\)](#)). [Eső and Szentes \(2007a\)](#) analyze how a seller should reveal information to bidders in an auction and show that a handicap auction can implement the revenue-maximizing information disclosure policy. [Bergemann and Bonatti \(2015\)](#) studies how a data broker sells consumer tracking information (cookies) to advertisers, and how the sale of that information affects advertising strategies and market outcomes. In their paper, the price of the information is determined in a competitive market, and the data buyer did not have any private information. [Bergemann et al. \(2015\)](#) characterize the welfare limits of price discrimination and show that market segmentations can attain any division of surplus between the monopolist and consumers, provided that the monopolist earns at least her uniform-price profit, and consumer surplus is non-negative.

Methodologically, this paper contributes to the literature on multidimensional screening ([Rochet and Stole, 2003](#); [Daskalakis et al., 2017](#)). The data broker screens over the set of all signals rather than a scalar quality, and the resulting type space is not totally ordered. Standard screening relies on a sorting condition: types are ranked so that the high type's incremental value of a better allocation exceeds the low type's. That fails here even with two types, because the value of a supplemental signal depends on the signal the decision maker already owns. The same difficulty arises in the work of [Haghpanah and Siegel \(2025\)](#), who consider screening with arbitrary allocation space and unordered types, but restrict attention to two types. In contrast, the screening problem in this paper involves arbitrary type spaces. What makes it tractable is the condition on payoffs that I identify: each type can be offered a signal worth more to her than to any other type, and the construction works for any type space.

3 Model

A decision maker (*she*) faces uncertainty about the true state of the world and acquires supplemental information from a data broker (*he*). The state of the world ω is drawn from a finite set $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$. The decision maker chooses an action a from the set of actions $A = \{a_1, a_2, \dots, a_m\}$. The payoff function given the action and the state is given by

$$u : A \times \Omega \rightarrow \mathbb{R}.$$

The payoff function can also be represented by the following matrix:

$u(a_i, \omega_j)$	ω_1	ω_2	$\cdots$	ω_n
a_1	u_{11}	u_{12}	$\cdots$	u_{1n}
a_2	u_{21}	u_{22}	$\cdots$	u_{2n}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
a_m	u_{m1}	u_{m2}	$\cdots$	u_{mn}

Assume that there is a unique optimal action in each state. That is,

$$\forall \omega \in \Omega, \exists a^*(\omega) \in A, u(a^*(\omega), \omega) > u(a, \omega), \forall a \in A, a \neq a^*(\omega).$$

Let $\mu \in \Delta\Omega$ be the common prior on Ω . For any belief $\mu \in \Delta(\Omega)$, the decision maker takes an optimal action $a(\mu)$ that solves

$$\max_{a \in A} \sum_{\omega \in \Omega} \mu(\omega) u(a, \omega).$$

Besides the prior, the decision maker privately holds additional information that enables her to make better decisions.

3.1 Information

I model the information as a signal π , following [Gentzkow and Kamenica \(2017\)](#). A *signal* is a finite partition of the extended state space $\Omega \times X$, where X is a random variable, independent of ω and uniformly distributed on $[0, 1]$. Let S denote the collection of nonempty Lebesgue-measurable subsets of $\Omega \times X$, and write $\pi \subset S$. An element $s \in \pi$ is called a *message* (or a *signal realization*). Define

$$\mu(s \mid \omega) = \lambda\{x \mid (\omega, x) \in s\} \quad \text{and} \quad \mu(s) = \sum_{\omega \in \Omega} \mu(s \mid \omega) \mu(\omega),$$

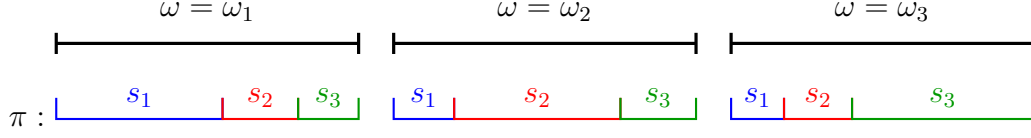


Figure 3: A signal.

where $\lambda(\cdot)$ denotes Lebesgue measure. For any $\omega \in \Omega$ define

$$s(\omega) = \{x \in [0, 1] \mid (\omega, x) \in s\}.$$

Moreover, $\mu(s \mid \omega)$ denotes the conditional probability of receiving s in state ω , and $\mu(s)$ denotes the unconditional probability of receiving s . For each $\omega \in \Omega$, the sets $\{s(\omega)\}_{s \in \pi}$ form a partition of $[0, 1]$. That is,

$$\sum_{s \in \pi} \mu(s \mid \omega) = 1 \quad \forall \omega \in \Omega. \quad (1)$$

Let Π denote the set of all signals, and let $\bar{\pi} = \{(\{\omega\} \times [0, 1]) \mid \omega \in \Omega\}$ denote the fully revealing signal. A signal captures how observations from one information source correlate with observations from others. This provides an algebraic structure over the set of all signals that allows us to add signals and analyze their joint information content (Brooks et al., 2024).

3.1.1 Value of supplemental information

The value of supplemental information depends on the information it augments. If the decision maker owns a signal π , then for each message $s \in \pi$, she will update her prior belief μ and choose an action $a(s) \in A^*(s)$, where

$$A^*(s) = \arg \max_{a \in A} \sum_{\omega \in \Omega} \mu(\omega \mid s) u(a, \omega). \quad (2)$$

Her interim utility conditional on receiving a message s is

$$u(s) = \sum_{\omega \in \Omega} \mu(\omega \mid s) u(a(s), \omega). \quad (3)$$

Her expected utility from a signal π is therefore given by

$$U(\pi) = \sum_{s \in \pi} \mu(s) u(s), \quad (4)$$

Her expected utility from always taking the optimal action is ²

$$\bar{U} = \sum_{\omega \in \Omega} \mu(\omega) u(a^*(\omega), \omega). \quad (5)$$

Next, given any two signals $\pi, \pi' \in \Pi$, define their *join* $\pi \vee \pi'$ by

$$\pi \vee \pi' = \{s \cap s' \mid s \in \pi, s' \in \pi', \mu(s \cap s') > 0\}. \quad (6)$$

That is, $\pi \vee \pi'$ is the signal that gives the same information as observing the two signals π and π' . The join to two signals is illustrated in Figure 4. The expected utility of $\pi \vee \pi'$ is

$$U(\pi \vee \pi') = \sum_{s \in \pi} \sum_{s' \in \pi'} \mu(s \cap s') u(s \cap s'). \quad (7)$$

The net value from adding π' to π is

$$V(\pi' \mid \pi) = U(\pi' \vee \pi) - U(\pi). \quad (8)$$

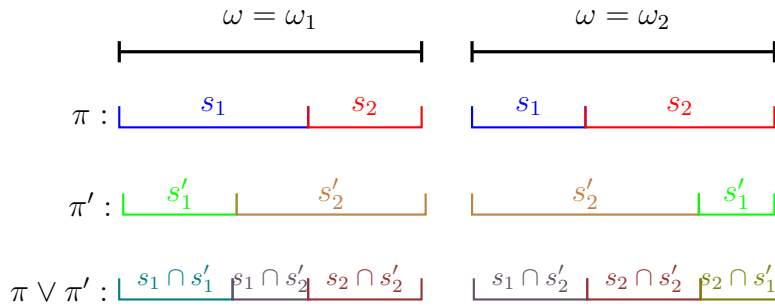


Figure 4: The join of two signals.

²Note that the decision maker does not necessarily need a fully revealing signal to always take the optimal action. To see this, consider three states and two actions, where the first action is optimal in the first two states and the second action is optimal in the third. A signal that separates state 3 from the first two states does not fully reveal the state, yet it still induces the optimal action in every state.

3.2 Data broker's problem

The data broker does not observe the decision maker's signal and therefore screens the decision maker based on the signal she already has. I assume that the marginal cost of providing additional information is zero. This assumption captures settings in which the data broker owns a massive database and the costs of packaging and distributing it are negligible.

By direct revelation principle, the data broker can restrict attention to direct mechanisms $\mathcal{M} = \{(\sigma(\pi), t(\pi))\}_{\pi \in \mathcal{T}}$ that assign a signal $\sigma(\pi)$ and a price $t(\pi)$ to each type π of decision maker. Let $\mathcal{T} \subseteq \Pi$ denote the decision maker's type space, with types distributed according to F on $\mathcal{T}$. Timing is as follows:

- (i) The data broker posts a menu $\mathcal{M} = \{(\sigma(\pi), t(\pi))\}_{\pi \in \mathcal{T}}$ of signal–price pairs before the true state is realized.
- (ii) A decision maker of type π purchases a signal $\sigma(\pi)$ and pays $t(\pi)$, before her private message is realized.
- (iii) The decision maker observes a signal s from π and another signal s' from $\sigma(\pi)$ and chooses an action $a \in A$.

Define

$$V(\pi) \triangleq V(\sigma(\pi) \mid \pi) = U(\sigma(\pi) \vee \pi) - U(\pi),$$

to be the net benefit of type π from reporting truthfully. The data broker chooses a menu to maximize expected revenue

$$\int_{\mathcal{T}} t(\pi) dF(\pi), \tag{9}$$

subject to incentive compatibility constraints

$$V(\pi) - t(\pi) \geq V(\sigma(\pi') \mid \pi) - t(\pi'), \quad \forall \pi, \pi' \in \mathcal{T}, \tag{IC}$$

and participation constraints

$$V(\pi) - t(\pi) \geq 0, \quad \forall \pi \in \mathcal{T}. \tag{IR}$$

The data broker's problem departs from the classic screening problems e.g., ([Mussa and Rosen, 1978](#); [Myerson, 1981](#); [Maskin and Riley, 1984](#)) in two ways. First, the object that is being allocated is multidimensional: an alternative is a signal and not a quality or quantity. Second, different types rank signals differently. Hence, the single-crossing property fails.

4 Analysis

In this section, I analyze the data brokers' problem: I provide conditions under which data broker extracts the full surplus, and provide an explicit construction of the optimal menu. A menu that extracts the full surplus assigns to each type a *confirm-or-reveal* signal. When full surplus extraction fails, I characterize the optimal menu where the decision maker has two types.

First, I will provide conditions under which the data broker can extract the full surplus. Take any finite type space $\mathcal{T} \subset \Pi$, and define for any $\pi \in \mathcal{T}$,

$$\underline{V}(\pi) = \min_{\tilde{\pi} \in \Pi} \max_{\pi' \in \mathcal{T}} V(\tilde{\pi} \mid \pi') \quad \text{subject to} \quad U(\tilde{\pi} \vee \pi) = \bar{U}. \quad (10)$$

The goal of solving problem (10) for a type π is to identify a signal that induces the optimal action for π in every state while minimizing the largest benefit that another type obtains from it. The following proposition shows that $\underline{V}(\pi)$ is well defined for every $\pi \in \mathcal{T}$.

Proposition 1. *$\underline{V}(\pi)$ is well defined for any $\pi \in \mathcal{T}$.*

The proof is in Appendix 6.1. Solutions of problem (10) will be crucial to construct the optimal menu that extracts the full surplus.

Definition 1. *Let $\mathcal{T}$ be finite. The data broker extracts the full surplus from the decision maker if there exists a feasible menu*

$$\mathcal{M}^* = \{(\sigma^*(\pi), t^*(\pi))\}_{\pi \in \mathcal{T}}$$

such that, for every $\pi \in \mathcal{T}$, $U(\sigma^(\pi) \vee \pi) = \bar{U}$, and the transfer satisfies $t^*(\pi) = \bar{U} - U(\pi)$.*

Proposition 2. *Let $\mathcal{T}$ be a finite set of types and F a distribution over $\mathcal{T}$. The data broker extracts the full surplus if and only if*

$$\underline{V}(\pi) \leq \bar{U} - U(\pi) \quad \text{for all } \pi \in \mathcal{T}.$$

The proof is in Appendix 6.2. The condition does not depend on the distribution of the type space, so whether the data broker can extract the full surplus depends only on what information types have.

4.1 Binary actions

Let $A = \{a_1, a_2\}$ and $\Omega = \{\omega_1, \dots, \omega_n\}$. The payoff function $u : A \times \Omega \rightarrow \mathbb{R}$ can be written as follows

$u(a, \omega)$	ω_1	ω_2	$\cdots$	ω_n
a_1	u_{11}	u_{12}	$\cdots$	u_{1n}
a_2	u_{21}	u_{22}	$\cdots$	u_{2n}

Observe that for any type $\pi \in \mathcal{T}$ and any $\sigma_\pi \in \Pi$ with $U(\sigma_\pi \vee \pi) = \bar{U}$, σ_π requires separating the event on which π induces optimal suboptimal actions. Let $R(\pi)$ denote the event on which π induces the decision maker to take optimal actions, and $W(\pi)$ denote the event on which the decision maker takes suboptimal actions following π . Formally, for any $\pi \in \Pi$, let $s_\pi(\omega, x)$ denote the message of π containing (ω, x) , and define

$$R(\pi) = \{(\omega, x) : a(s_\pi(\omega, x)) = a^*(\omega)\}, \quad W(\pi) = (\Omega \times [0, 1]) \setminus R(\pi).$$

Indeed, $a(s_\pi(\omega, x))$ may not be unique, in which case we can choose an arbitrary action from the set of optimal actions $A^*(s)$ for every message s . This pin down $R(\pi)$ for any π . The results are not affected by this choice. Furthermore, define

$$\mathcal{C}(\pi) = \{(\{R(\pi)\} \cup P) \setminus \{\emptyset\} : P \text{ is a finite partition of } W(\pi)\}.$$

The signals in $\mathcal{C}(\pi)$ are composed of messages that either confirm an optimal action or, combined with π 's message, identify the correct one. I call $\mathcal{C}(\pi)$ to be the set of all *confirm-or-reveal* signals of π . Every signal $\pi^c \in \mathcal{C}(\pi)$ complements π in the sense that $\pi \vee \pi^c$ always induces an optimal action. The following lemma shows that for any $\pi \in \mathcal{T}$, any signal in $\mathcal{C}(\pi)$ is a solution to (10). Moreover, all signals in $\mathcal{C}(\pi)$ are more worth to π than they are to any other type, which is what the following lemma shows.

Lemma 1. *Assume $|A| = 2$. Then, for all $\pi \in \mathcal{T}$, $\underline{V}(\pi) = \bar{U} - U(\pi)$, and every $\pi^c \in \mathcal{C}(\pi)$ solves (10).*

The proof is in Appendix 6.3. Every signal $\pi^c \in \mathcal{C}(\pi)$ consists of one message confirming that the decision maker's current action is optimal, together with an arbitrary partition of the event on which it is not. The coarsest one, $\pi^c = \{R(\pi), W(\pi)\}$, uses a single message on each, and is precisely a *switch/don't-switch* recommendation: the decision maker knows the action induced by her own signal, so the message $R(\pi)$ tells her not to switch, while $W(\pi)$ tells her to switch, which with two actions identifies the optimal action. Finer signals split $W(\pi)$ and so disclose more about the state, but only where π is inducing a suboptimal

action, so they leave her willingness to pay unchanged. With more than two actions, $W(\pi)$ no longer identifies the optimal action. Section 4.2 imposes a refinement of P under which each message induces the decision maker to take an optimal action.

Lemma 1 shows that the condition for full surplus extraction in Proposition 2 hold. The following proposition formalizes this result.

Proposition 3. *Assume $\Omega = \{\omega_1, \dots, \omega_n\}$ and $A = \{a_1, a_2\}$. The data broker extracts the full surplus from the decision maker. In particular, a menu*

$$\mathcal{M} = \{(\sigma(\pi), t(\pi)) \mid \pi \in \mathcal{T}\}$$

is optimal whenever $\sigma(\pi) \in \mathcal{C}(\pi)$ and $t(\pi) = \bar{U} - U(\pi)$ for every $\pi \in \mathcal{T}$.

The proof is in Appendix 6.4. Below I will illustrate the intuition in a simple example. Consider two states, good and bad, i.e., $\Omega = \{G, B\}$, and two actions $A = \{g, b\}$. Without loss of generality, normalize $u_{12} = u_{21} = 0$, $u_{22} = 1$, and $u_{11} = x > 0$. The payoffs are given by the following matrix

	G	B
g	x	0
b	0	1

Take any two types $\pi, \pi' \in \Pi$ and let s_1 be the union of all messages of π that induce action g , and s_2 the union of all messages of π that induce action b . Also let $\pi^c = \{s, ns\}$ be a *switch-don't switch* signal for π , as shown in Figure 5. That is, π^c sends message s whenever the message sent by π induces a suboptimal action, and ns whenever the message sent by π induces the optimal action. The net benefit of adding π^c to π is given by

$$V(\pi^c \mid \pi) = \mu(G, s_2) \cdot x + \mu(B, s_1) \cdot 1.$$

For simplicity, assume that π' consists of a single message s' . The same intuition extends to an arbitrary signal. Without loss of generality, assume that the optimal action under s' is b . A decision maker who has π' can mimic π by either obeying or disobeying the recommendation of π^c . The net benefit from obeying the recommendation of π^c is

$$\mu(G, s_2) x - \mu(B, s_1) \cdot 1.$$

This is clearly no greater than $V(\pi^c \mid \pi)$. When receiving the “do not switch” message, neither type gains any benefit. When receiving the “switch” message, type π always improves her action, whereas type π' takes the correct action with some probability and the wrong

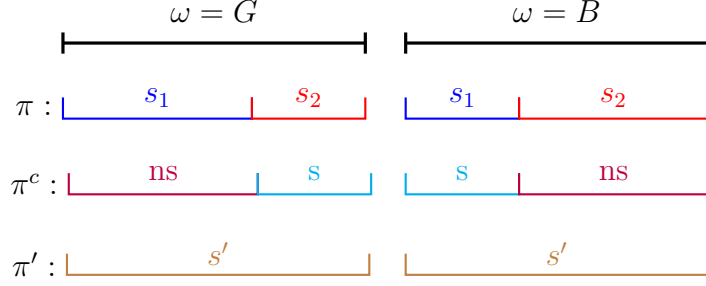


Figure 5: Intuition of Lemma 1.

one with the remaining probability.

On the other hand, type π' 's net benefit from disobeying the recommendation of π^c is given by

$$\mu(G \cap s_1) x - \mu(B \cap s_2) = [\mu(G) x - \mu(B)] + [\mu(B \cap s_1) - \mu(G \cap s_2) x].$$

Disobeying the “switch/don’t-switch” recommendation is equivalent to first changing the default action from b to g , and then obeying the recommendation as if g were the default. First, switching the default action from b to g leads to a loss, since before receiving any information, taking action b yields a higher expected payoff. Second, we have already shown that the benefit from obeying is no greater than $V(\pi^c | \pi)$.

Since π' is taken to be uninformative here, the pair of signals (π, π^c) are complements in this decision problem, in the sense of [Börger et al. \(2013\)](#): the switch/don’t-switch signal is worth more to the type it was built for than to an uninformed decision maker. Their notion fixes the pair of signals and requires this in every decision problem; incentive compatibility instead fixes the decision problem and requires the comparison against every signal $\pi' \in \Pi$, not only the uninformative one.

4.2 Many states and many actions

The data broker may not always extract the full surplus beyond the binary action case. To show this, I first construct a counterexample in which the data broker cannot extract the full surplus. Then, I analyze the environment with many states and actions, and provide sufficient conditions on the payoff structure under which data broker extracts the full surplus.

4.2.1 When full surplus extraction fails

The following example illustrates the case in which the data broker cannot extract the full surplus from the decision maker. Suppose the decision maker is a monopolist seller who sells a single indivisible good to consumers with valuations in $V = \{v_1, v_2, v_3\}$, with $v_1 < v_2 < v_3$ that follows some distribution $\mu \in \Delta(V)$. Assume that, for this given prior μ , the monopolist is indifferent between charging any price $p \in \{v_1, v_2, v_3\}$. That is,

$$v_1 = (\mu_2 + \mu_3)v_2 = \mu_3 v_3.$$

To distinguish actions (prices) from states (consumer valuations), let p_i denote the prices. The monopolist's payoff function is then represented by the following matrix:

	v_1	v_2	v_3
p_1	v_1	v_1	v_1
p_2	0	v_2	v_2
p_3	0	0	v_3

In this setting, the marginal benefit from being correct in the second or third state does not only depend on the consumer valuation, but also on the price the monopolist is setting. Consider two types of the monopolist $\mathcal{T} = \{\pi_L, \pi_H\}$, where $\pi_H = V \times [0, 1]$ is the uninformative signal, and $\pi_L = \{\{v_1, v_2\} \times [0, 1], \{v_3\} \times [0, 1]\}$ is a signal that pools states v_1 and v_2 together and reveals when the state is v_3 (see Figure 6). Type π_H will be referred as the "high type" because she has the highest willingness to pay for the full information, and type π_L as the low type.

Type π_H charges a uniform price, say p_1 (she is indifferent between charging any price) and earns a profit $U(\pi_H) = v_1$. Type π_L charges $p_3 = v_3$ to v_3 consumers, and $p_1 = v_1$ to both v_1 and v_2 consumers, since $v_1 = (\mu_2 + \mu_3)v_2$, implies $(\mu_1 + \mu_2)v_1 \geq \mu_2 v_2$. We can think of signals as action recommendations, that is think of π_H as having only one message p_1 , which indicates "charge $p_1 = v_1$ " and signal π_L as having two messages p'_1, p'_3 which recommends charging $p'_1 = v_1$ when consumer is known to have valuations v_1 or v_2 , and recommends charging $p'_3 = v_3$ where it is known that the consumer has valuation v_3 . To show that the full surplus extraction does not hold in this setting, let's compute $\underline{V}(\pi_L)$ and show that $\underline{V}(\pi_L) > \bar{U} - U(\pi_L)$. Let $\pi^m(\pi_L)$ be the solution to the following problem (i.e., the least valuable signal for π_H that induces correct action to π_L).

$$\min_{\tilde{\pi} \in \Pi} V(\tilde{\pi} \mid \pi_H) \quad \text{subject to} \quad U(\tilde{\pi} \vee \pi_L) = \bar{U}. \quad (11)$$

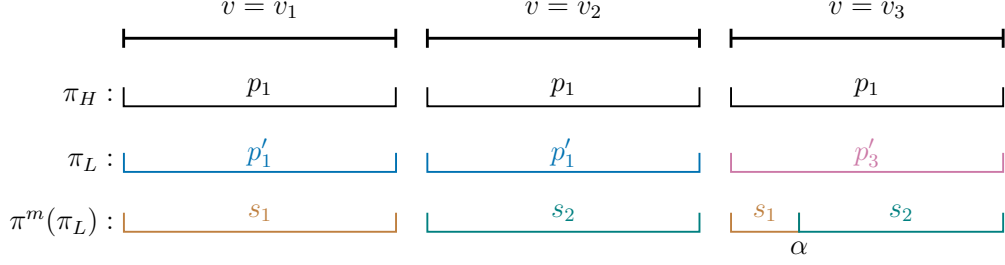


Figure 6: Candidate $\pi^m(\pi_L)$ with cutoff α .

Since every action is optimal in only one state, $\pi^m(\pi_L) \vee \pi_L$ should reveal the true state for $U(\pi^m(\pi_L) \vee \pi_L) = \bar{U}$. Thus, $\pi^m(\pi_L)$ separates v_1 consumers from v_2 consumers i.e., $\mu(s | v_1) > 0$ implies $\mu(s | v_2) = 0$ for every $s \in \pi^m(\pi_L)$. Moreover, it is weakly optimal to consider only two messages $s_1, s_2 \in \pi^m(\pi_L)$ with $s_1(v_1) = [0, 1]$ and $s_2(v_2) = [0, 1]$, because if there are more we can merge all messages with $s(v_1) > 0$ together and all messages with $s(v_2) > 0$ together and still have $U(\pi^m(\pi_L) \vee \pi_L) = \bar{U}$. Note that $\pi^m(\pi_L)$ becomes (weakly) less valuable for the high type whenever we merge some of its messages. So, it is without loss now to have $\pi^m(\pi_L) = \{s_1, s_2\}$ where $s_1 = (\{v_1\} \times [0, 1]) \cup (\{v_3\} \times [0, \alpha])$ and $s_2 = (\{v_2\} \times [0, 1]) \cup (\{v_3\} \times [\alpha, 1])$ for some $\alpha \in [0, 1]$. The solution is $\alpha^* = \frac{\mu_1 v_1}{\mu_3(v_3 - v_1)}$, and the derivation is in Appendix 6.5. Hence $\pi^m(\pi_L)$ with $\alpha^* = \frac{\mu_1 v_1}{\mu_3(v_3 - v_1)} \in (0, 1)$ solves (11).

Low type's benefit from $\pi^m(\pi_L)$ is

$$V(\pi^m(\pi_L) | \pi_L) = \mu(v_2)(v_2 - v_1).$$

The high type is indifferent between charging $p = v_1$ or $p = v_3$ after receiving signal realization s_1 . So, she benefits from $\pi^m(\pi_L)$ when she changes her price for v_2 and v_3 consumers, from charging them $p_1 = v_1$ to charging $p_2 = v_2$. That is,

$$V(\pi^m(\pi_L) | \pi_H) = \mu(v_2)(v_2 - v_1) + \mu(v_3, s_2)(v_2 - v_1) > V(\pi^m(\pi_L) | \pi_L).$$

since $\mu(v_3, s_2) > 0$. Finally, for any $\tilde{\pi}$ with $U(\tilde{\pi} \vee \pi_L) = \bar{U}$,

$$\max_{\pi' \in \mathcal{T}} V(\tilde{\pi} | \pi') \geq V(\tilde{\pi} | \pi_H) \geq V(\pi^m(\pi_L) | \pi_H) > \bar{U} - U(\pi_L),$$

hence $\underline{V}(\pi_L) > \bar{U} - U(\pi_L)$ and, by Proposition 2, the data broker cannot extract the full surplus in this environment.

Observe that whenever consumer valuation is v_3 the low type ignores message s_2 . Nevertheless, π_H benefits from it in that case. In particular, when $v = v_3$ and $s = s_2$, she improves her action, although the improvement comes from taking two different suboptimal actions,

that is, moving from charging the price $p = v_1$ to charging a price $p = v_2$.

4.2.2 Sufficient conditions for full surplus extraction

Consider $|\Omega| \geq 3$, $|A| \leq |\Omega|$ and the payoff function $u : A \times \Omega \rightarrow \mathbb{R}$ to be such that

$$\forall \omega \in \Omega, \quad u(a, \omega) = u(a', \omega) \quad \forall a, a' \in A \setminus \{a^*(\omega)\}. \quad (12)$$

That is, all suboptimal actions yield the same payoff in any given state, as illustrated below.

$u(a_i, \omega_j)$	ω_1	ω_{21}	ω_{22}	$\cdots$	ω_n
a_1	u_1	u'_{21}	u'_{22}	$\cdots$	u'_n
a_2	u'_1	u_{21}	u_{22}	$\cdots$	u'_n
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
a_m	u'_1	u'_{21}	u'_{22}	$\cdots$	u_n

Under this payoff structure, the marginal value of being corrected depends only on the state, not on the action the decision maker would otherwise take. Many matching problems have this form: routing a package to the correct address, serving the ad for the product a user actually wants, or matching a fingerprint to the right individual. In each case a wrong action is simply wrong, and the loss does not depend on which wrong action was made.

4.2.3 Illustrative example: Clinical diagnosis

A physician faces three conditions $\Omega = \{\omega_1, \omega_2, \omega_3\}$ (bacterial pneumonia, viral pneumonia, lung cancer), each with prior probability $1/3$. There are three treatments $A = \{a_1, a_2, a_3\}$ (antibiotics, antivirals, chemotherapy). Payoffs are diagonal:

$$u(a_i, \omega_j) = \begin{cases} 10, & i = j, \\ 0, & i \neq j. \end{cases}$$

A bacterial culture test perfectly identifies bacterial pneumonia, whereas imaging does not distinguish viral pneumonia from early-stage lung cancer (both can produce similar opacities). The resulting partition is $\{\{\omega_1\}, \{\omega_2, \omega_3\}\}$. When the culture is positive, the physician prescribes a_1 (antibiotics). When imaging is ambiguous, she chooses one treatment—say a_2 (antivirals)—which is correct in ω_2 and incorrect in ω_3 . The resulting expected utility is $10/3 + 10/3 + 0 = 20/3$. Without any additional information, prescribing a_1 yields $10/3$.

Suppose a private laboratory can sell any diagnostic test but is uncertain whether the

physician can run the culture test or not. The lab can offer a menu of tests that consist of a full screening test i.e., $\{\{\omega_1\}, \{\omega_2\}, \{\omega_3\}\}$ for $\$20/3$, and a test that can only screen for lung cancer and result in a partition of $\{\{\omega_1, \omega_2\}, \{\omega_3\}\}$ for $\$10/3$.

For the buyer who owns a bacterial culture test, the test that screen for lung cancer can be thought as a confirm-or-reveal signal, because it will confirm that the action is correct whenever the lung cancer result is negative, and will reveal when it is positive.

Incentives are as follows. For the informed type, the recommendation induces full information and increases expected utility from $20/3$ to 10 , so the incremental value is $10/3$; paying $10/3$ leaves zero rent. Purchasing the recommendation of the uninformed type, priced at $20/3$ is unprofitable. For the uninformed type, who would otherwise prescribe antibiotics, the recommendation also yields 10 i.e., an incremental value of $20/3$; paying $20/3$ leaves zero rent. Moreover, the uninformed type does not benefit by mimicking the informed type: that signal corrects only the mistakes defined relative to the informed type's baseline: it reveals in ω_3 but not in ω_2 , yielding at most the same incremental value $10/3$.

4.2.4 Confirm-or-reveal signals with many actions

With more than two actions, learning that her action is suboptimal no longer tells the decision maker what the optimal action is. Recall the even in which the decision maker takes a suboptimal action $W(\pi)$. For each $a \in A$ let

$$W_a(\pi) = \{(\omega, x) \in W(\pi) : a^*(\omega) = a\},$$

be the event on which π induces a suboptimal action and a is the correct action, and redefine the class of Section 4.1 as

$$\mathcal{C}(\pi) = \{(\{R(\pi)\} \cup P) \setminus \{\emptyset\} : P \text{ a finite partition of } W(\pi) \text{ refining } \{W_a(\pi)\}_{a \in A}\}.$$

Its coarsest element $\hat{\pi} = \{R(\pi)\} \cup \{W_a(\pi)\}_{a \in A}$ confirms that the decision maker's action is correct whenever the interim belief induced by π leads her to the optimal action, and reveals the correct action otherwise. Thus, every element $\pi^c \in \mathcal{C}(\pi)$ induces the correct action when combined with π i.e., $U(\pi^c \vee \pi) = \bar{U}$.

Lemma 2. Assume that $\Omega = \{\omega_1, \dots, \omega_n\}$, $A = \{a_1, \dots, a_m\}$ with $m \leq n$, and $u(a, \omega)$ satisfies (12). Then, for any $\pi \in \mathcal{T}$, $\underline{V}(\pi) = \bar{U} - U(\pi)$, and every $\pi^c \in \mathcal{C}(\pi)$ solves (10).

The proof is in Appendix 6.6. The intuition is as follows. Consider two types: an *informed* type, who owns a signal π , and an *uninformed* type, who takes a default action $a' \in A$. Let $\hat{\pi} = \{R(\pi)\} \cup \{W_a(\pi)\}_{a \in A} \in \mathcal{C}(\pi)$ be the coarsest *confirm-or-reveal* signal of π .

The informed type benefits from $\hat{\pi}$ exactly when she receives a message $W_a(\pi)$ (i.e., on the event where π induces a suboptimal action). That is, her valuation for $\hat{\pi}$ is

$$V_I(\hat{\pi}) = \sum_{\omega \in \Omega} \mu(W(\pi), \omega) \Delta(\omega),$$

where $\Delta(\omega)$ is the marginal gain from being corrected in state ω .

Denote $\bar{a}$ the action that the uninformed type takes after observing $R(\pi)$. Consider a dummy type who is committed to $\bar{a}$ before observing $\hat{\pi}$. The informed type changes her action is all the events where the dummy type changes, and on those events the marginal gain from being corrected is the same for both, since it depends only on the state. Thus, dummy type's gain cannot exceed $V_I(\hat{\pi})$. On the other hand, uninformed type's gain cannot exceed the dummy type's gain since taking action a' is optimal. Therefore,

$$V_U(\hat{\pi}) = U(\hat{\pi}) - u(a') \leq U(\hat{\pi}) - u(\bar{a}) \leq \sum_{a \in A} \sum_{\omega \in \Omega} \mu(W_a(\pi), \omega) \Delta(\omega) = V_I(\hat{\pi}).$$

Proposition 4. Assume $\Omega = \{\omega_1, \dots, \omega_n\}$, $A = \{a_1, \dots, a_m\}$ with $m \leq n$, $u(a, \omega)$ satisfies (12). The data broker extracts the full surplus from the decision maker. In particular, a menu

$$\mathcal{M} = \{(\sigma(\pi), t(\pi)) \mid \pi \in \mathcal{T}\}$$

is optimal whenever $\sigma(\pi) \in \mathcal{C}(\pi)$ and $t(\pi) = \bar{U} - U(\pi)$ for every $\pi \in \mathcal{T}$.

The proof is in appendix 6.7. Any mismatch between the chosen action and the realized state yields the same baseline payoff. As a result, the marginal value of being corrected depends only on the state and is independent of the action the decision maker would otherwise take. By contrast, under general payoff structures, the marginal value of being corrected typically depends on the default action the decision maker is taking, which is what makes the full surplus extraction fail.

4.2.5 Ruling out actions

The menus designed using signals in $\mathcal{C}(\pi)$ are not the only ones that extract the full surplus, hence assigning a *confirm-or-reveal signal* to each type is not the unique way to extract the full surplus. Indeed, for some confirm-or-reveal signal relative to π , one can merge the messages that reveal some of the actions, such that when combined with the messages of π it leaves one action standing. In that case, that message is interpreted as a message that rules out the actions that are not optimal. In other words, there is a larger class of signals that the data broker can use to extract the full surplus; signals whose messages either confirm an

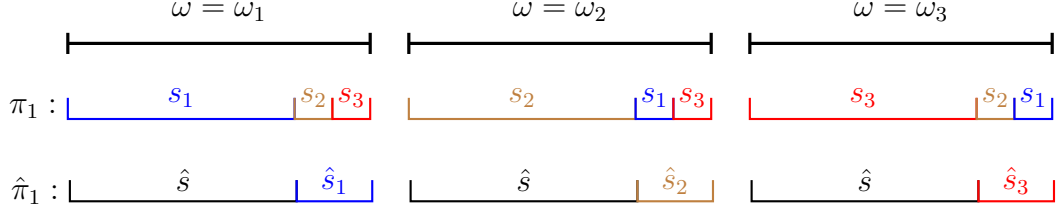


Figure 7: An example where there is no signal obtained from $\hat{\pi}_1$ by merging its messages such that it reveals the true state when combined with π_1 .

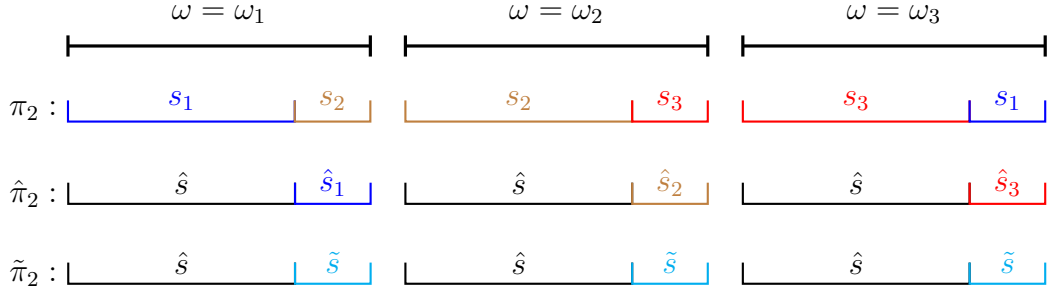


Figure 8: An example where merging messages of $\hat{\pi}_2$ yields a strictly less informative signal that still reveals the true state when combined with π_2 .

optimal action, or rule out actions. It may be that for some signals in $\mathcal{C}(\pi)$ we cannot do such a merging. How far merging can go depends on π , as the next two examples show.

First, consider the signal π_1 and its coarsest confirm-or-reveal signal $\hat{\pi}_1$, illustrated in Figure 7. Note that any signal obtained from $\hat{\pi}_1$ by merging its messages does not fully reveal the correct action when combined with π_1 . Thus, no coarsening of $\hat{\pi}_1$ reveals the true state when combined with π_1 .

Second, consider the signal π_2 and its coarsest confirm-or-reveal signal $\hat{\pi}_2$, illustrated in Figure 8. Construct $\tilde{\pi}_2$ from $\hat{\pi}_2$ by merging signals $\hat{s}_1, \hat{s}_2, \hat{s}_3$ together in one message $\tilde{s}$. That is $\tilde{\pi}_2 = \{\tilde{s}, \hat{s}\}$. In this case, $\tilde{\pi}_2$ reveals the correct action when combined with π_2 and is Blackwell dominated by $\hat{\pi}_2$. Nevertheless, no coarsening of $\tilde{\pi}_2$ reveals the true state when combined with π_2 .

4.3 Two types

Consider $\mathcal{T} = \{\pi_L, \pi_H\}$, with $U(\pi_L) > U(\pi_H)$; i.e., $\bar{V}_H = \bar{U} - U(\pi_H) > \bar{U} - U(\pi_L) = \bar{V}_L$. The high type is the one with the higher willingness to pay for full information. Let $\mu_H \in (0, 1)$

be the data broker's prior that the decision maker is the high type. The data broker solves

$$\begin{aligned}
& \max_{((\sigma_H, t_H), (\sigma_L, t_L))} \mu_H t_H + (1 - \mu_H) t_L \\
& \text{subject to} \\
& (IC_H) : \quad V_H(\sigma_H) - t_H \geq V_H(\sigma_L) - t_L, \\
& (IC_L) : \quad V_L(\sigma_L) - t_L \geq V_L(\sigma_H) - t_H, \\
& (IR_H) : \quad V_H(\sigma_H) - t_H \geq 0, \\
& (IR_L) : \quad V_L(\sigma_L) - t_L \geq 0,
\end{aligned} \tag{P}$$

where $V_i(\sigma_j) = V(\sigma_j \mid \pi_i)$ for $i, j \in \{L, H\}$.

Proposition 5. *The optimal menu $(\sigma_L, t_L), (\sigma_H, t_H)$ is such that*

- a. The high type always takes the optimal action i.e., $U(\sigma_H \vee \pi_H) = \bar{U}$.*
- b. IR_L binds.*
- c. If the full surplus extraction is not possible, then IC_H binds.*

The proof is provided in Appendix 6.8 and it follows an argument similar to that in Haghanah and Siegel (2025). Let σ_H^* be such that $V_H(\sigma_H^*) = \bar{V}_H$, Proposition 5 reduces the problem to

$$\begin{aligned}
& \max_{\sigma_L, t_H} \mu_H t_H + (1 - \mu_H) V_L(\sigma_L) \\
& \text{subject to} \\
& (IC_H) : \quad \bar{V}_H - t_H \geq V_H(\sigma_L) - V_L(\sigma_L), \\
& (IR_H) : \quad \bar{V}_H - t_H \geq 0, \\
& (IC_L) : \quad V_L(\sigma_H^*) - t_H \leq 0.
\end{aligned}$$

When full surplus extraction is not possible, from Proposition 5(c), the data broker sets $t_H = \bar{V}_H - V_H(\sigma_L) + V_L(\sigma_L)$, and (IR_H) becomes $V_H(\sigma_L) \geq V_L(\sigma_L)$. The problem reduces further to only choosing a σ_L^* that solves

$$\begin{aligned}
& \max_{\sigma_L \in \Pi} V_L(\sigma_L) - \mu_H V_H(\sigma_L) \\
& \text{subject to} \\
& (IR_H) : \quad V_H(\sigma_L) \geq V_L(\sigma_L), \\
& (IC_L) : \quad \bar{V}_H - V_L(\sigma_H^*) \geq V_H(\sigma_L) - V_L(\sigma_L).
\end{aligned} \tag{13}$$

Note that $(1 - \mu_H)V_L(\sigma_L)$ is the revenue the data broker extracts from the low type and $\mu_H(V_H(\sigma_L) - V_L(\sigma_L))$ is the rent he must leave the high type to keep her from mimicking the low type. The data broker therefore faces a trade-off between the two, and what matters about σ_L is only the pair of values the two types get from σ_L . This suggests fixing the low type's value and asking how small the high type's value can be made subject to it. The following lemma shows that the set of all pairs of values the two types get from σ_L is convex.

Lemma 3. *The set $S = \{(V_L(\sigma), V_H(\sigma)) \mid \sigma \in \Pi\}$ is convex.*

The proof is in appendix 6.9. Now, for any $z \in [0, \bar{V}_L]$, define

$$f(z) = \min_{\sigma \in \Pi} V_H(\sigma) \quad \text{subject to} \quad V_L(\sigma) = z, (IR_H). \quad (14)$$

Note that any pair $(z, f(z))$ satisfies (IC_L) .³ Then, having solved for f ,⁴ the problem (13) is transformed into finding $z^* \in [0, \bar{V}_L]$ that solves

$$\max_{z \in [0, \bar{V}_L]} z - \mu_H \cdot f(z). \quad (15)$$

The following proposition characterizes when the optimal solution is only excluding or inducing full information to the low type.

Proposition 6. *Assume the data broker cannot extract the full surplus. Then, f is convex. Excluding the low type is optimal if and only if $\mu_H f'(0^+) \geq 1$. In particular, the low type is served for every $\mu_H \in (0, 1)$ if and only if $f'(0^+) \leq 1$. Excluding the low type is never optimal even when $\mu_H \rightarrow 1$, whenever types are such that, there exists an event $E \subseteq \Omega \times [0, 1]$ where the low type takes a suboptimal action, while the high type takes a unique correct action (i.e., there exist $\omega \in \Omega$, $s_L \in \pi_L$, $s_H \in \pi_H$ with $\lambda(s_L(\omega) \cap s_H(\omega)) > 0$, such that $a^*(\omega)$ is the unique optimal action induced by s_H and $a(s_L) \neq a^*(\omega)$).*

Proof is in Appendix 6.10. The following examples illustrate the two cases the proposition 6 considers: In Example 1, f is linear with $f'(0^+) > 1$, so the objective is linear in z and the solution is bang-bang i.e., the low type either receives full information or is excluded. In Example 2, $f(z) = z$ on a neighborhood around zero, so $f'(0^+) = 1$ and the low type is served for every $\mu_H \in [0, 1]$.

³By Lemma 3, the line connecting $(0, 0)$ and $(\bar{V}_L, f(\bar{V}_L))$ lies in S , and each of its points satisfies (IR_H) since $f(\bar{V}_L) > \bar{V}_L$ when full surplus extraction fails. Hence $f(z) \leq \frac{f(\bar{V}_L)}{\bar{V}_L} z$ for $z \leq \bar{V}_L$. Finally, because $f(\bar{V}_L) \leq \bar{V}_H$, we have $f(z) - z \leq \bar{V}_H - \bar{V}_L \leq \bar{V}_H - V_L(\sigma_H^*)$.

⁴A minimizer σ_L^* exists using a similar argument as in the proof of Proposition 1.

4.3.1 Example 1: linear f

We now return to the monopolist example from Section 4.2 with type space $\mathcal{T} = \{\pi_L, \pi_H\}$. Recall that, $\pi_H = V \times [0, 1]$ is the uninformative signal, and $\pi_L = \{\{v_1, v_2\} \times [0, 1], \{v_3\} \times [0, 1]\}$ is a signal that pools states v_1 and v_2 together and reveals when the state is v_3 . The common prior μ is such that charging any price is optimal, hence $v_1 = (\mu_2 + \mu_3)v_2 = \mu_3 v_3$. Before computing $f(z)$, I first show that it is without loss to restrict attention to signals σ_L with at most two messages, in the sense that the solution to (13) lies in this class for every μ_H . In particular, letting $\pi_L = \{s'_1, s'_2\}$, where $s'_1 = \{v_1, v_2\} \times [0, 1]$ and $s'_2 = \{v_3\} \times [0, 1]$, the signal $\sigma_L = \{m_1, m_2\}$ is such that $s'_1 \cap m_1$ induces the decision maker to charge v_1 and $s'_1 \cap m_2$ induces her to charge v_2 . When $v = v_3$, she charges v_3 regardless of whether she receives m_1 or m_2 (see Figure 9).

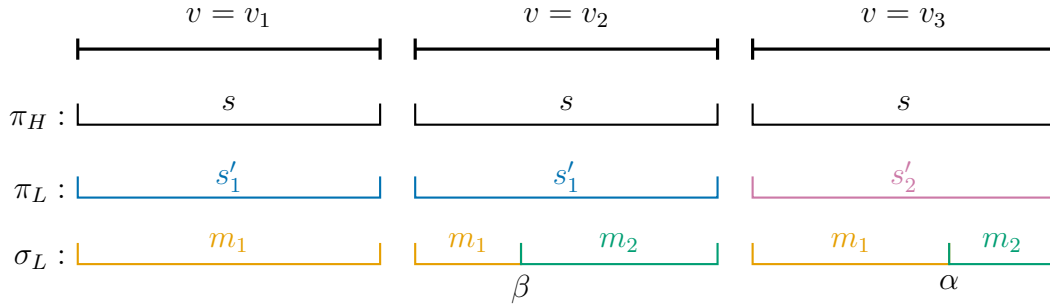


Figure 9: Candidate σ_L with cutoffs α and β .

First, note that it is optimal that $m_1(v_1) = 1$. Otherwise, we can transfer some mass from $m_2(v_1)$ to $m_1(v_1)$ so that V_L will increase by a $\varepsilon \cdot \mu_1 \cdot v_1$. On the other hand, V_H will increase by at most $\varepsilon \cdot \mu_1 \cdot v_1$. So $V_L - \mu_H \cdot V_H$ will increase for any μ_H .

Second, m_2 should induce price v_2 . Suppose instead that m_2 strictly induced price v_3 . Then m_1 must strictly induce v_1 or v_2 (or indifferent between v_1 and v_2), since otherwise v_3 would be strictly optimal at both messages and hence the unique default price for the high type, contradicting that every price is optimal under the prior. Now move some mass from $m_2(v_3)$ to $m_1(v_3)$. The low type charges v_3 in state v_3 after either message, so V_L is unchanged. The high type now charges those consumers v_1 or v_2 instead of v_3 , so V_H decreases.

Third, m_1 should make the high type indifferent between charging v_1 and v_3 . If m_1 strictly induced v_2 , then v_2 would be the unique default price, since m_2 induces v_2 by the previous step. Now, suppose that m_1 strictly induced price v_1 . Then shift a small mass ε from $m_1(v_2)$ to $m_2(v_2)$, which leaves the price at m_1 unchanged. This mass now sells at v_2 rather than v_1 , so both V_L and V_H increase by $\mu_2 \cdot \varepsilon \cdot (v_2 - v_1)$, and hence $V_L - \mu_H V_H$ increases

by $\mu_2 \cdot \varepsilon \cdot (v_2 - v_1)(1 - \mu_H) > 0$. Suppose instead that m_1 strictly induced price v_3 . Then m_2 must strictly induce v_2 and not v_3 , since otherwise charging v_3 would be the unique default price for the high type. Shifting a small mass from $m_1(v_3)$ to $m_2(v_3)$ therefore leaves V_L unchanged while V_H decreases.

We can therefore restrict attention to signals $\sigma_L = \{m_1, m_2\}$ with $m_1(v_1) = [0, 1]$, $m_1(v_2) = [0, \beta]$, and $m_1(v_3) = [0, \alpha]$ for some $\alpha, \beta \in [0, 1]$ (see Figure 9).

Then, we can compute $f(z)$. The low type's benefit from $\sigma = \{m_1, m_2\}$ is

$$V_L(\sigma) = \mu_2 \cdot (1 - \beta) \cdot (v_2 - v_1) = z.$$

The high type benefits from σ only when she receives message m_2 . In this case, she changes her price from v_1 to v_2 . Hence,

$$V_H(\sigma) = \underbrace{\mu_2 \cdot (1 - \beta) \cdot (v_2 - v_1)}_z + \mu_3 \cdot (1 - \alpha) \cdot (v_2 - v_1).$$

Then, since the high type is indifferent between charging v_1 or v_3 at signal m_1 , we have

$$\mu_1 v_1 + \mu_2 \beta v_1 + \mu_3 \alpha v_1 = \alpha \mu_3 v_3.$$

This gives,

$$\mu_1 v_1 + \mu_2 v_1 + \mu_3 v_1 - (1 - \beta) \mu_2 v_1 - (1 - \alpha) \mu_3 v_1 = \mu_3 v_3 - (1 - \alpha) \mu_3 v_3.$$

Using the fact that without any additional information all prices are optimal, we have $v_1 = \mu_3 v_3$. So, the expression gives

$$(1 - \alpha) \mu_3 (v_3 - v_1) = (1 - \beta) \mu_2 v_1.$$

This gives,

$$V_H(\sigma) = z + \mu_3 (1 - \alpha) (v_2 - v_1) = z + \frac{(1 - \beta) \mu_2 (v_2 - v_1) v_1}{v_3 - v_1} = z + \frac{z \cdot v_1}{v_3 - v_1} = \frac{v_3}{v_3 - v_1} z.$$

That is,

$$f(z) = \frac{v_3}{v_3 - v_1} z.$$

which is linear in z . Therefore, if $\mu_H < \frac{v_3 - v_1}{v_3}$, the low type ends up taking the correct action, and the high type gets positive rents. If $\mu_H > \frac{v_3 - v_1}{v_3}$, then only the high type is served. At $\mu_H = \frac{v_3 - v_1}{v_3}$ the objective is constant in z , so every allocation of the low type is optimal.

4.3.2 Example 2: the low type is always served

Consider a modified version of Example 1. Let the type space be $\mathcal{T} = \{\pi_L, \pi_H\}$ where $\pi_H = V \times [0, 1]$ is the uninformative signal, and $\pi_L = \{\{v_1, v_2\} \times [0, 1], \{v_3\} \times [0, 1]\}$ is a signal that pools states v_1 and v_2 together and reveals when the state is v_3 . Let $\mu \in \Delta(V)$ be such that

$$(\mu_2 + \mu_3)v_2 > \max\{v_1, \mu_3 v_3\}, \quad (16)$$

$$\frac{v_3 - v_2}{v_3 - v_1} \mu_1 v_1 > \mu_2 (v_2 - v_1). \quad (17)$$

From (16), type π_H charges the unique optimal price v_2 . From (17), the data broker cannot extract the full surplus. To see this, note that $\sigma_L = \pi^m(\pi_L)$ with $\alpha = \frac{\mu_1 v_1}{\mu_3(v_3 - v_1)}$ as in Figure 6 is a solution to $\underline{V}(\pi_L)$, which follows the same logic as in Section 4.2. The left-hand side of equation (17) is $\underline{V}(\pi_L)$ and the right-hand side is $\bar{U} - U(\pi_L)$.⁵

Since v_2 is the unique optimal price under the prior, there exists $\varepsilon_0 > 0$ such that $\sigma_L = \{s_1, s_2\}$ with $s_2 = (\{v_2\} \times [0, \varepsilon_0])$, $s_1 = (V \times [0, 1]) \setminus s_2$ such that $V_L(\sigma_L) = \mu_2 \varepsilon_0 (v_2 - v_1) = z_0$. Since charging v_2 is uniquely optimal for the high type, we can choose ε_0 sufficiently small such that the high type's action remains unchanged after both messages. Thus, $V_H(\sigma_L) = 0$. So, $(z_0, 0) \in S$. Since $(0, 0) \in S$ and since S is convex, we have $(z, 0) \in S$ for all $z \leq z_0$. Moreover, since both $(z, 0)$, $(z, \frac{\bar{V}_H}{\bar{V}_L} z) \in S$ ⁶, by convexity of S , we have $(z, z) \in S$ for every $z \leq z_0$. That is, $f(z) = z$ for all $z \leq z_0$, implying that $f'(0^+) = 1$. Therefore, low type is always served even when the frequency of the high type μ_H approaches 1.

5 Conclusion

I study how a data broker sells information, in the form of a signal, to a decision maker who privately owns a different signal. The decision maker faces uncertainty about a payoff-relevant state and purchases information at the ex ante stage. This captures situations in which the decision maker makes frequent decisions, such as charging prices to different types of consumers, or approving and rejecting loan applications. The data broker does not observe the signal the decision maker owns, and therefore screens her over the space of all signals.

In standard screening problems the seller degrades quality or excludes low types. Here, by contrast, the data broker extracts the full surplus in a large class of environments, by assigning to each type a *confirm-or-reveal* signal. I provide a necessary and sufficient condition for full surplus extraction, and show that it holds whenever the decision maker chooses

⁵(17) implies that $\mu_1 v_1 > \mu_2 (v_2 - v_1)$. Thus, low type charges price v_1 to both v_1 and v_2 consumers. Hence the low type's benefit from σ_L is exactly $\mu_2 (v_2 - v_1)$.

⁶ $(z, \frac{\bar{V}_H}{\bar{V}_L} z) \in S$ because it's an element of the line that connects $(0, 0)$ and $(\bar{V}_L, \bar{V}_H) \in S$.

between two actions, and, more generally, whenever all suboptimal actions yield the same payoff in any given state. When the condition fails, I characterize the optimal menu with two types, and give a condition on the two types under which the low type is served for any prior of the data broker.

The data broker can sell signals that are arbitrarily correlated with the one the decision maker already owns, and this plays a central role in the design of the optimal menu. It enables him to design the supplemental information such that each decision maker type ends up taking the correct action even after purchasing a partially informative signal, and it is what allows him to leave no rents although he does not observe what the decision maker already holds. The menu that does so is deterministic: signals are offered at fixed prices, and each type's transfer depends only on its own report. Full surplus extraction here therefore requires neither correlation across several agents' private types, as in [Cr  mer and McLean \(1988\)](#) and [McAfee and Reny \(1992\)](#), nor randomization over information structures, as in [Kr  hmer \(2020\)](#).

The framework in this paper captures situations where firms buy data using data append, that is by adding new characteristics to existing datasets. It also captures situations where data is sold sequentially but the contract is signed at the ex-ante stage; for example, a lender signs a contract with a credit bureau and then acquires reports on individual applicants over the life of that contract, after each application arrives. Two restrictions seem worth relaxing. First, the data broker owns the fully revealing signal and can sell any signal at zero marginal cost; a constrained broker, who owns only part of the data, would face a smaller set of signals he can sell and may no longer be able to tailor a confirm-or-reveal signal to every type. Second, other than considering the data broker to maximize revenue, one can instead ask which menu is optimal subject to the constraint that the decision maker always takes the correct action.

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6 Appendix

6.1 Proof of Proposition 1

Proof. To prove that a solution to (10) always exists, I first show that it is without loss to restrict attention to signals with finitely many messages. Then, combining all types into a single grand signal, the objective function depends on the signal only through the measure of the intersections of its messages with those of the grand signal, in each state. This reduces the problem to minimizing a continuous function over a closed subset of a compact set.

Consider any finite type space $\mathcal{T} = \{\pi_1, \dots, \pi_T\}$ and pick any type $\pi_i \in \mathcal{T}$. For any signal $\pi \in \Pi$, define its message profile with respect to type π_i by

$$\varphi_{i,\pi} : \pi \rightarrow (A \cup \{\emptyset\})^{\pi_i}, \quad \varphi_{i,\pi}(s) = (a(r, s))_{r \in \pi_i},$$

where $a(r, s)$ is an optimal action at the posterior $\mu(\cdot \mid r \cap s)$ whenever $\mu(r \cap s) > 0$, and $a(r, s) = \emptyset$ otherwise. If more than one action is optimal, then take $a(r, s)$ to be the one with the lowest index.

Take any signal $\pi \in \Pi$, and construct $\hat{\pi}$ from π by merging messages in π that have the same message profile. That is, letting $S_\pi(m) = \bigcup \{s \in \pi \mid \varphi_{i,\pi}(s) = m\}$ for any $m \in \varphi_{i,\pi}(\pi)$,

$$\hat{\pi} = \{S_\pi(m) \mid m \in \varphi_{i,\pi}(\pi)\}.$$

Note that $U(\pi_i \vee \hat{\pi}) = U(\pi_i \vee \pi)$, since type π_i takes the same action after each message of $\pi_i \vee \hat{\pi}$ as she did after the messages of $\pi_i \vee \pi$ it merges. For any other type $\pi_j \in \mathcal{T}$, $\hat{\pi}$ is a coarsening of π , so $U(\pi_j \vee \hat{\pi}) \leq U(\pi_j \vee \pi)$. Merging preserves the constraint in (10) and weakly lowers the objective function. So, it is without loss to merge all messages that have the same message profile. Therefore, it is without loss to restrict attention to signals whose messages are indexed by the set of profiles $\mathcal{M}_i = (A \cup \{\emptyset\})^{\pi_i}$, and hence have at most $M_i = (|A| + 1)^{|\pi_i|}$ messages. Denote the set of such signals by $\Pi^{\mathcal{M}_i}$.

Let $\pi^* = \bigvee_{\pi' \in \mathcal{T}} \pi'$ be the grand signal obtained by combining all signals in the type space. Let $\mathcal{D} = \{(\omega, s^*) \in \Omega \times \pi^* \mid \mu(s^* \mid \omega) > 0\}$. For any $\pi \in \Pi^{\mathcal{M}_i}$, define $y_\pi : \mathcal{D} \rightarrow \Delta(\mathcal{M}_i)$ such that $y_\pi(\omega, s^*)(m) = \mu(s \mid \omega, s^*)$, where $s \in \pi$ is the unique message with $\varphi_{i,\pi}(s) = m$, and $y_\pi(\omega, s^*)(m) = 0$ otherwise. Finally, define

$$Y = \Delta(\mathcal{M}_i)^{\mathcal{D}}.$$

Conversely, each $y \in Y$ is the representation of some signal, call it π_y : within each $s^*(\omega)$, assign to each $m \in \mathcal{M}_i$ with $y(\omega', s^*)(m) > 0$ for some $(\omega', s^*) \in \mathcal{D}$ a measurable subset of measure $y(\omega, s^*)(m) \cdot \mu(s^* \mid \omega)$. Note that for this π_y , its message indexed by m need not have message profile m .

Now, I will show that for any signal π whose messages are indexed by $\mathcal{M}_i$, $U(\pi_j \vee \pi)$ depends on π only through its representation y_π , for any type $\pi_j \in \mathcal{T}$. Take $\pi_j \in \mathcal{T}$. Since π^* refines π_j , each $r \in \pi_j$ is a union of messages of π^* . We can write $\pi = \{s_m\}_{m \in \mathcal{M}'_i}$, where $\mathcal{M}'_i \subseteq \mathcal{M}_i$. For every $m \in \mathcal{M}'_i$ and $\omega \in \Omega$,

$$\mu(r \cap s_m \mid \omega) = \sum_{s^* \in \pi^*: s^* \subseteq r} y_\pi(\omega, s^*)(m) \cdot \mu(s^* \mid \omega).$$

Hence

$$\begin{aligned} U(\pi_j \vee \pi) &= \sum_{r \in \pi_j} \sum_{m \in \mathcal{M}'_i} \mu(r \cap s_m) \cdot u(r \cap s_m) \\ &= \sum_{r \in \pi_j} \sum_{m \in \mathcal{M}'_i} \max_{a \in A} \sum_{\omega \in \Omega} \mu(\omega) \mu(r \cap s_m \mid \omega) \cdot u(a, \omega) \\ &= \sum_{r \in \pi_j} \sum_{m \in \mathcal{M}'_i} \max_{a \in A} \left[\sum_{\omega \in \Omega} \mu(\omega) u(a, \omega) \sum_{s^* \subseteq r} y_\pi(\omega, s^*)(m) \cdot \mu(s^* \mid \omega) \right]. \end{aligned} \tag{18}$$

From the last identity we observe that $U(\pi_j \vee \pi)$ depends on π only through y_π , so the data broker's choice can be taken to be an element of Y .

$\underline{V}(\pi_i)$ can be written as

$$\underline{V}(\pi_i) = \min_{y \in Y} \max_{j \leq T} \left[U(\pi_j \vee \pi_y) - U(\pi_j) \right] \quad \text{subject to} \quad U(\pi_i \vee \pi_y) = \bar{U}.$$

From (18), the map $y \mapsto U(\pi_j \vee \pi_y)$ is continuous, and since there are finitely many types, the map $y \mapsto \max_{j \leq T} (U(\pi_j \vee \pi_y) - U(\pi_j))$ is also continuous. The set Y is a finite product of simplices, hence compact. The set $\{y \in Y \mid U(\pi_i \vee \pi_y) = \bar{U}\}$ is closed, since $y \mapsto U(\pi_i \vee \pi_y)$ is continuous, and hence compact. Moreover, let $\bar{y}$ be the representation of the signal obtained from $\bar{\pi}$ by merging messages with the same message profile. Then $\bar{y} \in \{y \in Y \mid U(\pi_i \vee \pi_y) = \bar{U}\}$. Finally, because $\{y \in Y \mid U(\pi_i \vee \pi_y) = \bar{U}\}$ is a non-empty compact set and the objective function is continuous, $\underline{V}(\pi_i)$ is well defined. □

6.2 Proof of Proposition 2

Proof. Assume

$$\underline{V}(\pi) \leq \bar{U} - U(\pi) \quad \text{for each } \pi \in \mathcal{T}.$$

Let σ_π be the minimizer of the min max problem, which exists by Proposition 1. Then, consider a menu $\mathcal{M} = \{(\sigma_\pi, t_\pi)\}$, where $t_\pi = \bar{U} - U(\pi)$ for all $\pi \in \mathcal{T}$. Clearly, IR constraints bind and each type gets zero rent. To check the IC constraints, note that for any $\pi, \hat{\pi} \in \mathcal{T}$,

$$V(\sigma_\pi \mid \hat{\pi}) \leq \max_{\pi' \in \mathcal{T}} V(\sigma_\pi \mid \pi') = \underline{V}(\pi) = \bar{U} - U(\pi) = t_\pi.$$

So, deviating to π' 's signal, gives $\hat{\pi}$ payoff of $V(\sigma_\pi \mid \hat{\pi}) - t_\pi \leq 0 = V(\sigma_{\hat{\pi}} \mid \hat{\pi}) - t_{\hat{\pi}}$.

To prove the "only if" part, assume by contradiction that data broker extracts the full surplus, and there is a type π for which

$$\underline{V}(\pi) > \bar{U} - U(\pi). \tag{19}$$

Since, full surplus is possible, there is a feasible menu $\mathcal{M} = \{(\sigma_\pi, t_\pi)\}$, where $U(\sigma_\pi \vee \pi) = \bar{U}$ and $t_\pi = \bar{U} - U(\pi)$ for all $\pi \in \mathcal{T}$. Because each type receives zero rent, IC constraints write

$$V(\sigma_\pi \mid \pi) \geq V(\sigma_\pi \mid \pi') \quad \text{for all } \pi' \in \mathcal{T}.$$

This implies

$$V(\sigma_\pi \mid \pi) \geq \max_{\pi' \in \mathcal{T}} V(\sigma_\pi \mid \pi').$$

Since $V(\sigma_\pi \mid \pi) = \bar{U} - U(\pi)$, it follows that

$$\bar{U} - U(\pi) \geq \max_{\pi' \in \mathcal{T}} V(\sigma_\pi \mid \pi') \geq \min_{\tilde{\pi} \in \Pi} \max_{\pi' \in \mathcal{T}} \{V(\tilde{\pi} \mid \pi') \mid U(\tilde{\pi} \vee \pi) = \bar{U}\} = \underline{V}(\pi),$$

which contradicts with (19). □

6.3 Proof of Lemma 1

Proof. For every $\omega \in \Omega$, write $u_1(\omega) \triangleq u(a_1, \omega)$ and $u_2(\omega) \triangleq u(a_2, \omega)$. Assume $\pi \in \mathcal{T}$. First, I will show that every $\pi^c \in \mathcal{C}(\pi)$ induces the optimal action when combined with π . Then, I will show that the value of any signal $\pi^c \in \mathcal{C}(\pi)$ is maximized when combined with π . The final step concludes that $\underline{V}(\pi) = \bar{U} - U(\pi)$.

Step 1. Every $\pi^c \in \mathcal{C}(\pi)$ satisfies $U(\pi^c \vee \pi) = \bar{U}$.

Assume the decision maker observes $s \in \pi$ and $s^c \in \pi^c$. If $s^c = R(\pi)$, then $a(s) = a^*(\omega)$

for every ω with $\mu(\omega \mid s \cap s^c) > 0$. So, $a(s)$ is optimal there. Otherwise $s^c \subseteq W(\pi)$, so $a^*(\omega) \neq a(s)$ for every such ω , and since $|A| = 2$ this leaves a single action, optimal throughout $s \cap s^c$. Hence the decision maker takes the optimal action at every message of $\pi^c \vee \pi$, and $U(\pi^c \vee \pi) = \bar{U}$.

Step 2. For any $\pi^c \in \mathcal{C}(\pi)$, $V(\pi^c \mid \pi) \geq V(\pi^c \mid \pi')$ for all $\pi' \in \Pi$.

We can write $\pi^c = \{R(\pi)\} \cup P$, where P is a finite partition of $W(\pi)$, and

$$W(\pi) = \{(\omega, x) \mid a(s_\pi(\omega, x)) \neq a^*(\omega)\}$$

is the event in which π induces a suboptimal action. For any $\omega \in \Omega$, let

$$\Delta(\omega) = |u_1(\omega) - u_2(\omega)|$$

denote the gain from being corrected in state ω . By Step 1, $U(\pi^c \vee \pi) = \bar{U}$, so

$$V(\pi^c \mid \pi) = \bar{U} - U(\pi) = \sum_{\omega \in \Omega} \mu(W(\pi), \omega) \Delta(\omega) = \sum_{p \in P} \sum_{\omega \in \Omega} \mu(p, \omega) \Delta(\omega). \quad (20)$$

The second equality holds because π induces optimal action exactly on $R(\pi)$, and the third equality holds because P is a finite partition of $W(\pi)$.

Take any $\pi' \in \Pi$.

$$\begin{aligned} V(\pi^c \mid \pi') &= U(\pi^c \vee \pi') - U(\pi') \\ &= \sum_{s \in \pi'} \sum_{s^c \in \pi^c} \mu(s \cap s^c) u(s \cap s^c) - \sum_{s \in \pi'} \mu(s) u(s) \\ &= \sum_{s \in \pi'} \mu(s) \left[\sum_{s^c \in \pi^c} \mu(s^c \mid s) u(s \cap s^c) - u(s) \right] \\ &= \sum_{s \in \pi'} \mu(s) V(\pi^c \mid s). \end{aligned} \quad (21)$$

Now, fix $s \in \pi'$ and let $\bar{a}(s)$ be the action induced by $R(\pi) \cap s$ (if this event is null, let $\bar{a}(s)$ be arbitrary). Note that

$$u(s) \geq \sum_{\omega \in \Omega} \mu(\omega \mid s) u(\bar{a}(s), \omega) = \sum_{s^c \in \pi^c} \mu(s^c \mid s) \sum_{\omega \in \Omega} \mu(\omega \mid s^c, s) u(\bar{a}(s), \omega).$$

Then,

$$\begin{aligned}
V(\pi^c \mid s) &\leq \sum_{s^c \in \pi^c} \mu(s^c \mid s) u(s \cap s^c) - \sum_{s^c \in \pi^c} \mu(s^c \mid s) \sum_{\omega \in \Omega} \mu(\omega \mid s^c, s) u(\bar{a}(s), \omega) \\
&= \sum_{s^c \in \pi^c} \sum_{\omega \in \Omega} \mu(s^c \mid s) \mu(\omega \mid s^c, s) u(a(s \cap s^c), \omega) - \sum_{s^c \in \pi^c} \mu(s^c \mid s) \sum_{\omega \in \Omega} \mu(\omega \mid s^c, s) u(\bar{a}(s), \omega) \\
&= \sum_{s^c \in \pi^c} \sum_{\omega \in \Omega} \mu(s^c \mid s) \mu(\omega \mid s^c, s) [u(a(s \cap s^c), \omega) - u(\bar{a}(s), \omega)]
\end{aligned}$$

The term $s^c = R(\pi)$ in the summation vanishes, since $a(R(\pi) \cap s) = \bar{a}(s)$. Similarly, the terms $s^c = p \in P$ with $a(p \cap s) = \bar{a}(s)$ vanish. For $s^c = p \in P$ with $a(p \cap s) \neq \bar{a}(s)$, we have

$$u(a(p \cap s), \omega) - u(\bar{a}(s), \omega) = \begin{cases} \Delta(\omega), & \text{if } a^*(\omega) = a(p \cap s), \\ -\Delta(\omega), & \text{if } a^*(\omega) = \bar{a}(s). \end{cases}$$

Therefore,

$$V(\pi^c \mid s) \leq \sum_{p \in P} \sum_{\omega \in \Omega} \mu(p \mid s) \mu(\omega \mid p, s) \Delta(\omega).$$

Multiplying by $\mu(s)$ and summing over $s \in \pi'$ gives

$$\begin{aligned}
V(\pi^c \mid \pi') &= \sum_{s \in \pi'} \mu(s) V(\pi^c \mid s) \\
&\leq \sum_{s \in \pi'} \mu(s) \sum_{p \in P} \sum_{\omega \in \Omega} \mu(p \mid s) \mu(\omega \mid p, s) \Delta(\omega) \\
&= \sum_{s \in \pi'} \sum_{p \in P} \sum_{\omega \in \Omega} \mu(\omega, p, s) \Delta(\omega).
\end{aligned}$$

We finally have

$$V(\pi^c \mid \pi') \leq \sum_{s \in \pi'} \sum_{p \in P} \sum_{\omega \in \Omega} \mu(\omega, p, s) \Delta(\omega) = \sum_{p \in P} \sum_{\omega \in \Omega} \mu(\omega, p) \Delta(\omega) = V(\pi^c \mid \pi),$$

where the first equality holds because π' partitions $p(\omega)$ for every p and ω , and the second comes from (20).

Step 3. Any $\pi^c \in \mathcal{C}(\pi)$ solves (10).

From Step 1, $U(\pi^c \vee \pi) = \bar{U}$, so π^c is feasible and $V(\pi^c \mid \pi) = \bar{U} - U(\pi)$. By Step 2, since $V(\pi^c \mid \pi) \geq V(\pi^c \mid \pi')$ for all $\pi' \in \Pi$, it follows that

$$\max_{\pi' \in \mathcal{T}} V(\pi^c \mid \pi') = V(\pi^c \mid \pi) = \bar{U} - U(\pi).$$

Moreover, for any feasible $\tilde{\pi}$, $V(\tilde{\pi} \mid \pi) = U(\tilde{\pi} \vee \pi) - U(\pi) = \bar{U} - U(\pi)$. Hence

$$\max_{\pi' \in \mathcal{T}} V(\tilde{\pi} \mid \pi') \geq V(\pi^c \mid \pi) = \bar{U} - U(\pi).$$

Therefore $\underline{V}(\pi) = V(\pi^c \mid \pi) = \bar{U} - U(\pi)$ and any $\pi^c \in \mathcal{C}(\pi)$ attains it. □

6.4 Proof of Proposition 3

Proof. From Lemma 1, $\underline{V}(\pi) \leq \bar{U} - U(\pi)$ for all $\pi \in \mathcal{T}$. Then, by Proposition 2, data broker extracts the full surplus. Consider a menu with $\sigma(\pi) \in \mathcal{C}(\pi)$ and $t(\pi) = \bar{U} - U(\pi)$. By Step 1 of Lemma 1, $U(\sigma(\pi) \vee \pi) = \bar{U}$, so each type attains $\bar{U}$ and pays her valuation for full information, which is the maximum revenue the data broker can obtain. Every type therefore receives zero rent. By Step 2 of the proof of Lemma 1,

$$V(\sigma(\pi') \mid \pi) \leq \bar{U} - U(\pi') = t(\pi') \quad \text{for all } \pi, \pi' \in \mathcal{T},$$

so (IC) holds. □

6.5 Derivation of α in Section 4.2

Recall that $\pi^m(\pi_L) = \{s_1, s_2\}$ with $s_1 = (\{v_1\} \times [0, 1]) \cup (\{v_3\} \times [0, \alpha])$ and $s_2 = (\{v_2\} \times [0, 1]) \cup (\{v_3\} \times [\alpha, 1])$. Since $\pi^m(\pi_L) \vee \pi_H = \pi^m(\pi_L)$, we have

$$\begin{aligned} V(\pi^m(\pi_L) \mid \pi_H) &= U(\pi^m(\pi_L)) - U(\pi_H) \\ &= \mu(s_1) \cdot u(s_1) + \mu(s_2) \cdot u(s_2) - v_1 \\ &= \max\{(\mu_1 + \alpha\mu_3)v_1, \alpha\mu_3v_3\} + \max\{(\mu_2 + (1 - \alpha)\mu_3)v_2, (1 - \alpha)\mu_3v_3\} - v_1 \end{aligned}$$

Thus, the problem 11 reduces to finding $\alpha \in [0, 1]$ that solves

$$\min_{\alpha \in [0, 1]} \left[\max\{(\mu_1 + \alpha\mu_3)v_1, \alpha\mu_3v_3\} + \max\{(\mu_2 + (1 - \alpha)\mu_3)v_2, (1 - \alpha)\mu_3v_3\} \right].$$

Since $(\mu_2 + \mu_3)v_2 = \mu_3v_3$, we have $(\mu_2 + (1 - \alpha)\mu_3)v_2 > (1 - \alpha)\mu_3v_3$ for any $\alpha > 0$. It is not optimal to have $(\mu_1 + \alpha\mu_3)v_1 < \alpha\mu_3v_3$. Otherwise, decreasing α by some $\varepsilon > 0$ decreases the first term by $\varepsilon\mu_3v_3$ and increases the second term by $\varepsilon\mu_3v_2$. Hence, $(\mu_1 + \alpha\mu_3)v_1 \geq \alpha\mu_3v_3$.

Optimality requires $(\mu_1 + \alpha\mu_3)v_1 = \alpha\mu_3v_3$. To see this, assume $(\mu_1 + \alpha\mu_3)v_1 > \alpha\mu_3v_3$. Then $\alpha < 1$. Increasing α by some $\varepsilon > 0$ increases the first term by $\varepsilon\mu_3v_1$ and decreases the

second term by $\varepsilon\mu_3v_2$. Thus, one optimal solution sets α so that $(\mu_1 + \alpha\mu_3)v_1 = \alpha\mu_3v_3$, i.e.,

$$\alpha = \frac{\mu_1v_1}{\mu_3(v_3 - v_1)} \in (0, 1),$$

since $v_1(\mu_1 + \mu_3) < \mu_3v_3$.

6.6 Proof of Lemma 2

Proof. For $\omega \in \Omega$ let $u(\omega) = u(a^*(\omega), \omega)$ for the payoff from the optimal action, and let $u'(\omega)$ denote the payoff from any suboptimal action, which is well defined by (12). Let

$$\Delta(\omega) = u(\omega) - u'(\omega) > 0$$

denote the gain from being corrected in state ω .

Step 1. Every $\pi^c \in \mathcal{C}(\pi)$ satisfies $U(\pi^c \vee \pi) = \bar{U}$.

Assume the decision maker observes $s \in \pi$ and $s^c \in \pi^c$. If $s^c = R(\pi)$, then $a(s) = a^*(\omega)$ for every ω with $\mu(\omega \mid s \cap R(\pi)) > 0$. So, $a(s)$ will remain the optimal action. If $s^c = p \in P$, then $p \subseteq W_a(\pi)$ for some $a \in A$, because P refines $\{W_a(\pi)\}_{a \in A}$. Then, conditional on receiving signals s and p , it will be revealed that the optimal action is a . Thus, the decision maker always takes the optimal action following $\pi^c \vee \pi$.

Step 2. For any $\pi^c \in \mathcal{C}(\pi)$, $V(\pi^c \mid \pi) \geq V(\pi^c \mid \pi')$ for all $\pi' \in \Pi$.

Let $\pi^c = \{R(\pi)\} \cup P$ for some refinement partition P of $\{W_a(\pi)\}_{a \in A}$. By Step 1, since $U(\pi^c \vee \pi) = \bar{U}$, we have

$$V(\pi^c \mid \pi) = \bar{U} - U(\pi) = \sum_{\omega \in \Omega} \mu(W(\pi), \omega) \Delta(\omega) = \sum_{p \in P} \sum_{\omega \in \Omega} \mu(p, \omega) \Delta(\omega). \quad (22)$$

The second equality holds because π induces optimal action exactly on $R(\pi)$, and the third equality holds because P is a refinement partition of $\{W_a(\pi)\}_{a \in A}$ and hence a finite partition of $W(\pi)$.

Take any $\pi' \in \Pi$. As in the equation (21) in the proof of Lemma 1,

$$V(\pi^c \mid \pi') = \sum_{s \in \pi'} \mu(s) V(\pi^c \mid s).$$

Fix $s \in \pi'$ and let $\bar{a}(s)$ be the action induced by $R(\pi) \cap s$ (if this event is null, let $\bar{a}(s) = a(s)$).

Since $u(s)$ is the maximum over A ,

$$u(s) \geq \sum_{\omega \in \Omega} \mu(\omega \mid s) u(\bar{a}(s), \omega) = \sum_{s^c \in \pi^c} \mu(s^c \mid s) \sum_{\omega \in \Omega} \mu(\omega \mid s^c, s) u(\bar{a}(s), \omega),$$

and therefore

$$\begin{aligned} V(\pi^c \mid s) &\leq \sum_{s^c \in \pi^c} \mu(s^c \mid s) u(s \cap s^c) - \sum_{s^c \in \pi^c} \mu(s^c \mid s) \sum_{\omega \in \Omega} \mu(\omega \mid s^c, s) u(\bar{a}(s), \omega) \\ &= \sum_{s^c \in \pi^c} \sum_{\omega \in \Omega} \mu(s^c \mid s) \mu(\omega \mid s^c, s) u(a(s \cap s^c), \omega) - \sum_{s^c \in \pi^c} \mu(s^c \mid s) \sum_{\omega \in \Omega} \mu(\omega \mid s^c, s) u(\bar{a}(s), \omega) \\ &= \sum_{s^c \in \pi^c} \sum_{\omega \in \Omega} \mu(s^c \mid s) \mu(\omega \mid s^c, s) [u(a(s \cap s^c), \omega) - u(\bar{a}(s), \omega)] \end{aligned}$$

The term $s^c = R(\pi)$ vanishes, since $a(R(\pi) \cap s) = \bar{a}(s)$. For $s^c = p \in P$, Step 1 gives $a(s \cap p) = a(p) = a^*(\omega)$ for every ω with $\mu(\omega \mid s \cap p) > 0$. Thus,

$$u(a(s \cap p), \omega) - u(\bar{a}(s), \omega) = \begin{cases} 0, & \text{if } a^*(\omega) = \bar{a}(s), \\ \Delta(\omega), & \text{otherwise.} \end{cases}$$

Therefore

$$V(\pi^c \mid s) \leq \sum_{p \in P} \sum_{\omega \in \Omega} \mu(p \mid s) \mu(\omega \mid p, s) \Delta(\omega).$$

Multiplying by $\mu(s)$, using $\mu(s)\mu(p \mid s)\mu(\omega \mid p, s) = \mu(\omega, p, s)$, and summing over $s \in \pi'$, gives

$$V(\pi^c \mid \pi') \leq \sum_{s \in \pi'} \sum_{p \in P} \sum_{\omega \in \Omega} \mu(\omega, p, s) \Delta(\omega) = \sum_{p \in P} \sum_{\omega \in \Omega} \mu(\omega, p) \Delta(\omega) = V(\pi^c \mid \pi),$$

where the first equality holds because π' partitions $p(\omega)$ for every p and ω , and the second is (22).

Step 3. Any $\pi^c \in \mathcal{C}(\pi)$ solves (10).

Identical to Step 3 in the proof of Lemma 1. □

6.7 Proof of Proposition 4

Proof. From Lemma 2, $\underline{V}(\pi) \leq \bar{U} - U(\pi)$ for all $\pi \in \mathcal{T}$. Thus the condition of Proposition 2 holds and the data broker extracts the full surplus. Consider a menu with $\sigma(\pi) \in \mathcal{C}(\pi)$ and $t(\pi) = \bar{U} - U(\pi)$. By Step 1 of the proof of Lemma 2, $U(\sigma(\pi) \vee \pi) = \bar{U}$, so each type attains $\bar{U}$ and pays her valuation for full information, which is the maximum revenue the

data broker can obtain. Every type therefore receives zero rent. By Step 2 of the proof of Lemma 2,

$$V(\sigma(\pi') \mid \pi) \leq \bar{U} - U(\pi') = t(\pi') \quad \text{for all } \pi, \pi' \in \mathcal{T},$$

so (IC) holds. $\square$

6.8 Proof of Proposition 5

Proof. Assume $\{(\sigma_H, t_H), (\sigma_L, t_L)\}$ is an optimal allocation.

- a. Assume by contradiction that $U(\sigma_H \vee \pi_H) < \bar{U}$. Take $\sigma'_H \in \Pi$ such that $U(\sigma'_H \vee \pi_H) = \bar{U}$, implying that $V_H(\sigma'_H) > V_H(\sigma_H)$.

Case 1. Assume IR_H binds i.e., $t_H = V_H(\sigma_H)$. Set $t'_H = V_H(\sigma'_H)$. Then, $t'_H > t_H$. The high type's rent is still zero, so IR_H and IC_H hold, and IR_L remains unchanged. Note that,

$$V_L(\sigma'_H) - t'_H < V_H(\sigma'_H) - t'_H = 0 \leq V_L(\sigma_L) - t_L, \quad (23)$$

since $V_H(\sigma'_H) = \bar{U} - U(\pi_H) > \bar{U} - U(\pi_L) \geq V_L(\sigma'_H)$, and the last inequality holds by IR_L . Then, (23) shows that IC_L holds and hence implies that $\{(\sigma'_H, t'_H), (\sigma_L, t_L)\}$ is feasible, and since $t'_H > t_H$ it attains a higher profit, which is a contradiction.

Case 2. Assume IR_H does not bind. Then, IC_H must bind. Otherwise, the data broker can attain a higher profit by raising t_H by a small amount. Take

$$t'_H = V_H(\sigma'_H) - (V_H(\sigma_L) - t_L).$$

IR_L does not change, and IC_H still holds since $V_H(\sigma'_H) - t'_H = V_H(\sigma_L) - t_L$. Moreover, from IC_H binding we have $V_H(\sigma_H) - t_H = V_H(\sigma_L) - t_L$, hence

$$V_H(\sigma'_H) - t'_H = V_H(\sigma_L) - t_L = V_H(\sigma_H) - t_H \geq 0,$$

implying that IR_H holds. If IC_L holds, then $\{(\sigma'_H, t'_H), (\sigma_L, t_L)\}$ is feasible and yields higher profits.⁷ If IC_L does not hold, then we have

$$V_L(\sigma_L) - t_L < V_L(\sigma'_H) - t'_H.$$

Note that $t'_H \geq t_L$ since $V_H(\sigma'_H) \geq V_H(\sigma_L)$. But then, selling the same allocation (σ'_H, t'_H) to both types yields higher profits.

⁷ $t'_H = V_H(\sigma'_H) - (V_H(\sigma_L) - t_L) > V_H(\sigma_H) - (V_H(\sigma_L) - t_L) = V_H(\sigma_H) - (V_H(\sigma_H) - t_H)$. The last equality comes from the fact that IC_H binds. That is, $t'_H > t_H$.

- b. By contradiction, assume that IR_L does not bind. Then IC_L binds, because if IC_L is slack, then the data broker can raise t_L by a small amount and achieve a higher profit. Also, IR_H binds, because if IR_H was also slack, the data broker can raise both t_L and t_H by the same amount. Then, it follows that

$$\begin{aligned}
V_L(\sigma_L) - t_L &= V_L(\sigma_H) - t_H \quad \text{since } IC_L \text{ binds,} \\
&= V_L(\sigma_H) - V_H(\sigma_H) \quad \text{since } IR_H \text{ binds,} \\
&= V_L(\sigma_H) - (\bar{U} - U(\pi_H)) \\
&< 0 \quad \text{since } V_L(\sigma_H) \leq \bar{U} - U(\pi_L) < \bar{U} - U(\pi_H).
\end{aligned}$$

This contradicts the supposition that IR_L does not bind.

- c. By contradiction, assume that IC_H does not bind. Then IR_H must bind (if neither binds, this cannot be optimal, since the data broker could raise the price t_H). Since IR_H binds, we have $V_H(\sigma_H) - t_H = 0$. Hence, the IC_H constraint writes

$$V_H(\sigma_L) - t_L < V_H(\sigma_H) - t_H = 0.$$

From part (b), we have $t_L = V_L(\sigma_L)$. Hence, $V_H(\sigma_L) < V_L(\sigma_L)$. Since the data broker cannot extract the full surplus, we have $U(\sigma_L \vee \pi_L) < \bar{U}$. Otherwise, both allocations would be efficient, and since IR_L and IR_H bind, the menu would extract the full surplus. Thus, $V_L(\sigma_L) < \bar{U} - U(\pi_L)$. But then, since IC_H is slack, consider giving the low type a better signal σ'_L that will improve her utility and increase t_L without violating the IC_H constraint. That is, we can choose σ'_L so that $V_H(\sigma_L)$ will increase by a very small amount.

More formally, let us first show that for any $\varepsilon > 0$, there exists $\sigma'_L \in \Pi$ such that $V_H(\sigma'_L) \leq V_H(\sigma_L) + \varepsilon$ and $V_L(\sigma'_L) > V_L(\sigma_L)$. Since $U(\sigma_L \vee \pi_L) < \bar{U}$, there exist $s \in \sigma_L$ and $s' \in \pi_L$ such that $s \cap s'$ induces a suboptimal action in some state ω . Then, obtain σ'_L from σ_L by refining s in a way that reveals the true state ω only on a very small subset in $(s' \cap s)(\omega)$. This region can be chosen arbitrarily small so that $V_H(\sigma'_L) \leq V_H(\sigma_L) + \varepsilon$.

Then, consider σ'_L obtained from σ_L as above, and let $t'_L = t_L + V_L(\sigma'_L) - V_L(\sigma_L)$. Note that IR_L , IR_H , and IC_L will not be violated. Moreover, we can choose $\varepsilon > 0$ such that it will not violate IC_H .

□

6.9 Proof of Lemma 3

Proof. Consider

$$S = \{(V_L(\sigma), V_H(\sigma)) \mid \sigma \in \Pi\}.$$

Take any two points $(V_L(\sigma_1), V_H(\sigma_1))$ and $(V_L(\sigma_2), V_H(\sigma_2))$ for some $\sigma_1, \sigma_2 \in \Pi$. Then for any $\alpha \in (0, 1)$, we can construct σ such that

$$V_i(\sigma) = \alpha V_i(\sigma_1) + (1 - \alpha) V_i(\sigma_2), \quad i = L, H.$$

To do this, first consider the signal $\pi = \pi_L \vee \pi_H \vee \sigma_1 \vee \sigma_2$. Take any $s \in \pi$. Since $s(\omega) \subseteq [0, 1]$, there exist disjoint $s_1(s), s_2(s)$ with $s_1(s) \cup s_2(s) = s$, and $\lambda(s_1(\omega)) = \alpha \cdot \lambda(s(\omega))$ for every $\omega \in \Omega$, since λ is a Lebesgue measure on $[0, 1]$. Denote $B = \bigcup_{s \in \pi} s_1(s)$. Then by construction, for every $s \in \pi$ and $\omega \in \Omega$,

$$\mu(s \cap B \mid \omega) = \lambda(s_1(s)(\omega)) = \alpha \cdot \lambda(s(\omega)) = \alpha \cdot \mu(s \mid \omega),$$

so that B is independent of the state and of every message of π . In particular

$$\mu(B) = \sum_{\omega \in \Omega} \mu(\omega) \cdot \mu(B \mid \omega) = \sum_{\omega \in \Omega} \sum_{s \in \pi} \mu(\omega) \cdot \mu(s \cap B \mid \omega) = \alpha.$$

Now, construct

$$\sigma = \{m_1 \cap B \mid m_1 \in \sigma_1\} \cup \{m_2 \cap B^c \mid m_2 \in \sigma_2\}.$$

Since $\{m_1 \cap B\}_{m_1 \in \sigma_1}$ partitions B and $\{m_2 \cap B^c\}_{m_2 \in \sigma_2}$ partitions B^c , we have $\sigma \in \Pi$. Fix $i \in \{L, H\}$. Take any $r \in \pi_i$ and any $m_1 \in \sigma_1$. Both r and m_1 are unions of messages of π , and so is $r \cap m_1$. So, for every $\omega \in \Omega$,

$$\mu(r \cap m_1 \cap B \mid \omega) = \sum_{s \in \pi: s \subseteq r \cap m_1} \mu(s \cap B \mid \omega) = \sum_{s \in \pi: s \subseteq r \cap m_1} \alpha \cdot \mu(s \mid \omega) = \alpha \cdot \mu(r \cap m_1 \mid \omega).$$

The last equality comes from the fact that $\{s \in \pi \mid s \subseteq r \cap m_1\}$ is a partition of $r \cap m_1$. Symmetrically we have

$$\mu(r \cap m_2 \cap B^c \mid \omega) = (1 - \alpha) \cdot \mu(r \cap m_2 \mid \omega)$$

for every $m_2 \in \sigma_2$. Moreover, for any $r \in \pi_i$, $\omega \in \Omega$ and $m_1 \in \sigma_1$,

$$\begin{aligned}\mu(\omega \mid r \cap m_1 \cap B) &= \frac{\mu(\omega) \cdot \mu(r \cap m_1 \cap B \mid \omega)}{\sum_{\omega' \in \Omega} \mu(\omega') \cdot \mu(r \cap m_1 \cap B \mid \omega')} = \frac{\alpha \mu(\omega) \cdot \mu(r \cap m_1 \mid \omega)}{\alpha \sum_{\omega' \in \Omega} \mu(\omega') \cdot \mu(r \cap m_1 \mid \omega')} \\ &= \mu(\omega \mid r \cap m_1).\end{aligned}$$

Similarly, for any $r \in \pi_i$, $\omega \in \Omega$, and $m_2 \in \sigma_2$,

$$\begin{aligned}\mu(\omega \mid r \cap m_2 \cap B^c) &= \frac{\mu(\omega) \cdot \mu(r \cap m_2 \cap B^c \mid \omega)}{\sum_{\omega' \in \Omega} \mu(\omega') \cdot \mu(r \cap m_2 \cap B^c \mid \omega')} = \frac{(1 - \alpha) \mu(\omega) \cdot \mu(r \cap m_2 \mid \omega)}{(1 - \alpha) \sum_{\omega' \in \Omega} \mu(\omega') \cdot \mu(r \cap m_2 \mid \omega')} \\ &= \mu(\omega \mid r \cap m_2).\end{aligned}$$

That is, $r \cap m_1 \cap B$ induces the same posterior belief as $r \cap m_1$ and $r \cap m_2 \cap B^c$ induces the same posterior belief as $r \cap m_2$. Every message of σ is either of the form $m_1 \cap B$ or $m_2 \cap B^c$ with $m_1 \in \sigma_1$ and $m_2 \in \sigma_2$. Thus,

$$\begin{aligned}U(\sigma \vee \pi_i) &= \sum_{r \in \pi_i} \sum_{m_1 \in \sigma_1} \mu(r \cap m_1 \cap B) \cdot \max_{a \in A} \sum_{\omega \in \Omega} \mu(\omega \mid r \cap m_1 \cap B) \cdot u(a, \omega) \\ &\quad + \sum_{r \in \pi_i} \sum_{m_2 \in \sigma_2} \mu(r \cap m_2 \cap B^c) \cdot \max_{a \in A} \sum_{\omega \in \Omega} \mu(\omega \mid r \cap m_2 \cap B^c) \cdot u(a, \omega) \\ &= \sum_{r \in \pi_i} \sum_{m_1 \in \sigma_1} \alpha \cdot \mu(r \cap m_1) \cdot \max_{a \in A} \sum_{\omega \in \Omega} \mu(\omega \mid r \cap m_1) \cdot u(a, \omega) \\ &\quad + \sum_{r \in \pi_i} \sum_{m_2 \in \sigma_2} (1 - \alpha) \cdot \mu(r \cap m_2) \cdot \max_{a \in A} \sum_{\omega \in \Omega} \mu(\omega \mid r \cap m_2) \cdot u(a, \omega) \\ &= \alpha \cdot U(\sigma_1 \vee \pi_i) + (1 - \alpha) \cdot U(\sigma_2 \vee \pi_i).\end{aligned}$$

Subtract $U(\pi_i)$ from both sides and we get

$$V(\sigma \mid \pi_i) = \alpha V(\sigma_1 \mid \pi_i) + (1 - \alpha) V(\sigma_2 \mid \pi_i).$$

□

6.10 Proof of Proposition 6

Proof. Step 1. The function $f(z)$ is convex.

From Lemma 3, the set $Z = \{(x, y) \in S \mid y \geq x\}$ is also convex. Take any $z_1, z_2 \in [0, \bar{V}_L]$ and $\alpha \in (0, 1)$. So, there exist $\sigma_1, \sigma_2 \in \Pi$ such that $(V_L(\sigma_1), V_H(\sigma_1)) = (z_1, f(z_1)) \in Z$ and $(V_L(\sigma_2), V_H(\sigma_2)) = (z_2, f(z_2)) \in Z$. From Lemma 3, there exist $\sigma \in \Pi$ such that

$$V_L(\sigma) = \alpha z_1 + (1 - \alpha) z_2, \quad V_H(\sigma) = \alpha f(z_1) + (1 - \alpha) f(z_2).$$

Since Z is convex, $(V_L(\sigma), V_H(\sigma)) \in Z$. Thus, $f(\alpha z_1 + (1 - \alpha)z_2)$ being the minimized value, it follows that

$$f(\alpha z_1 + (1 - \alpha)z_2) \leq V_H(\sigma) = \alpha f(z_1) + (1 - \alpha)f(z_2).$$

Hence f is convex.

Step 2. The low type is excluded if and only if $f'(0^+) \geq \frac{1}{\mu_H}$.

Denote $R(z) = z - \mu_H \cdot f(z)$. Since f is convex, R is concave. So, $z = 0$ is a maximizer if and only if $R'(0^+) \leq 0$. That is,

$$R'(0^+) \leq 0 \iff 1 - \mu_H \cdot f'(0^+) \leq 0 \iff f'(0^+) \geq \frac{1}{\mu_H} > 1.$$

Note that, if $f'(0^+) \leq 1$, then $R'(0^+) > 0$, hence $z = 0$ cannot be a maximizer, even when μ_H is close to 1.

Step 3. To show the second part of the proposition, assume there exist $\omega \in \Omega$, $s_L \in \pi_L$, $s_H \in \pi_H$ with $\lambda(s_L(\omega) \cap s_H(\omega)) > 0$, such that $a(s_L) \neq a^*(\omega)$ and $a^*(\omega)$ is the unique optimal action induced by s_H .

Construct $\sigma_L = \{s_1, s_2\}$ such that $s_1(\omega) \subseteq s_L(\omega) \cap s_H(\omega)$, $\lambda(s_1(\omega)) = \delta$, $s_1(\omega') = \emptyset$ for all $\omega' \neq \omega$, and $s_2 = (\Omega \times [0, 1]) \setminus s_1$. I choose $\delta > 0$ such that $a(s_H) = a(s_H \cap s_2)$, which is possible since $a(s_H)$ is strictly optimal conditional on s_H .

Then, $V_H(\sigma_L) = 0$, while $V_L(\sigma_L) \geq \mu(\omega) \cdot \delta \cdot (u(a^*(\omega), \omega) - u(a(s_L), \omega)) = z_0 > 0$. Since $(z_0, 0) \in S$ and $(0, 0) \in S$, convexity of S gives $(z, 0) \in S$ for all $z \leq z_0$, and $(z, \frac{\bar{V}_H}{\bar{V}_L} z) \in S$; since $\frac{\bar{V}_H}{\bar{V}_L} z \geq z$, convexity gives $(z, z) \in S$, so the smallest y with $(z, y) \in Z$ is $y = z$.

Hence $f(z) = z$ on $[0, z_0]$ and $f'(0^+) = 1$. By Step 2, $R'(0^+) = 1 - \mu_H > 0$, so $z = 0$ is not a maximizer and type π_L is never excluded for any $\mu_H \in (0, 1)$. $\square$